

Lesson 12 : Sharing, Publishing, and Presenting Data Visualizations.

Dr. Mahendra Kanojia

What you will learn ?

- a) Export dashboards as PDF/Image
- b) Use presentation mode in Tableau
- c) Publish dashboards to Tableau Public
- d) Share dashboards via Tableau Server/Online
- e) Enable user interaction on shared dashboards
- f) Manage permissions and access control
- g) Evaluate and optimize sharing strategies

Version Compatibility Note:

- **Tableau Desktop (Free Trial / Student / Public 2026.2):** Full native support for local exports (PDF, PNG, EMF), Presentation Mode, publishing to Tableau Public, and client-side interactions (Actions, Tooltip Vizizes).
- **Tableau Server / Tableau Cloud Specific Features (Parts D & F):** Direct enterprise publishing, project folder governance, and active directory user permissions require a paid Tableau Server/Cloud environment. We will execute local and Tableau Public equivalents while covering the exact enterprise steps.

Dataset: [Netflix Movies and TV Shows](#)

Kaggle link: <https://www.kaggle.com/datasets/shivamb/netflix-shows?resource=download>

A: Export Dashboards as PDF/Image

Objective: Build an executive Netflix overview dashboard and export high-resolution vector PDFs and raster images (.png / .emf) formatted for print briefings and slide decks.

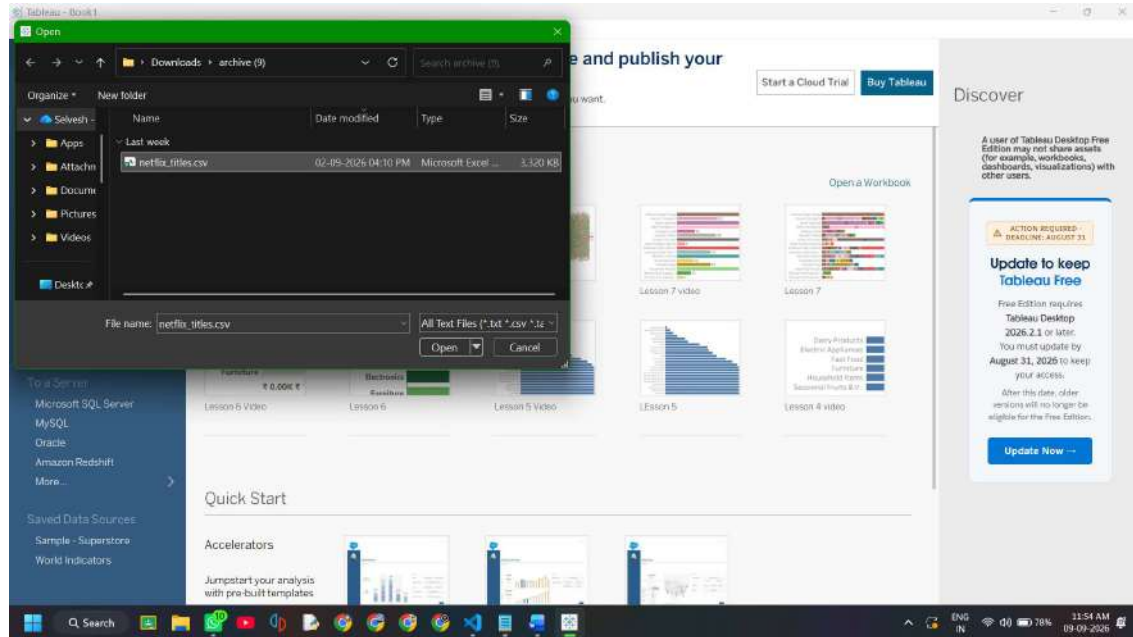
Stakeholders and executives often require static offline artifacts (board decks, email reports). Knowing how to configure print setups, page scaling, and vector exports ensures charts do not become cropped, blurry, or misaligned when exported outside of Tableau.

Step 1: Build Two Base Visualizations

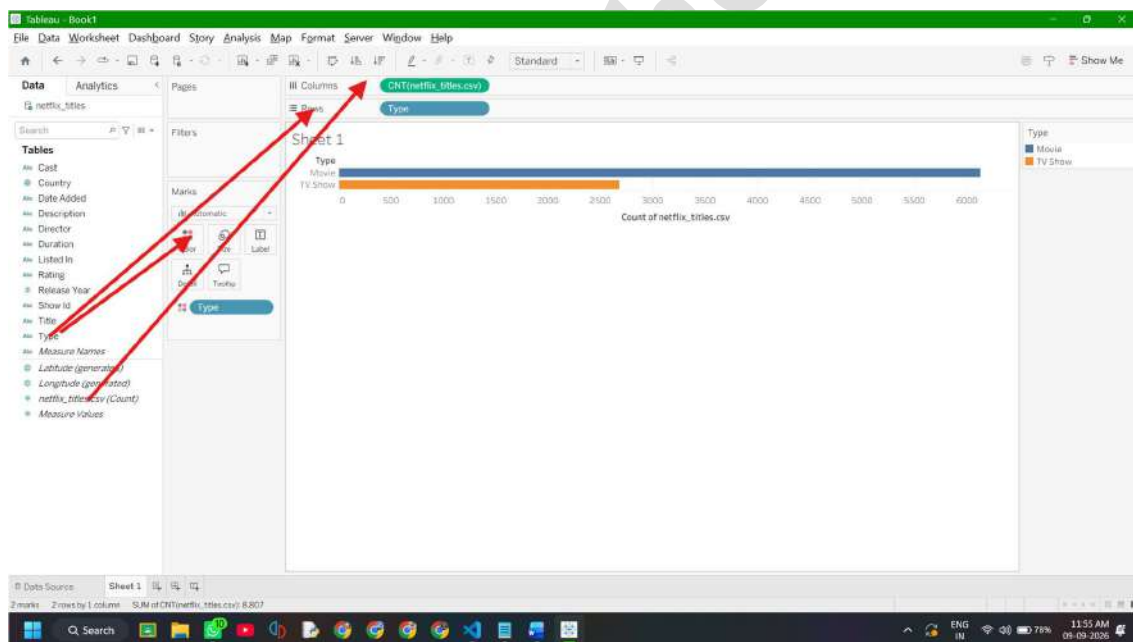
Sheet 1: Content Type Split (Bar Chart)

1. Connect to `netflix_titles.csv` (use Text file connection).

Data Visualization with Power BI and Tableau



2. Drag **type** to **Rows**.
3. Drag **netflix_titles.csv (Count)** (or number of records) to **Columns**.
4. Drag **type** to **Color** (Marks card).



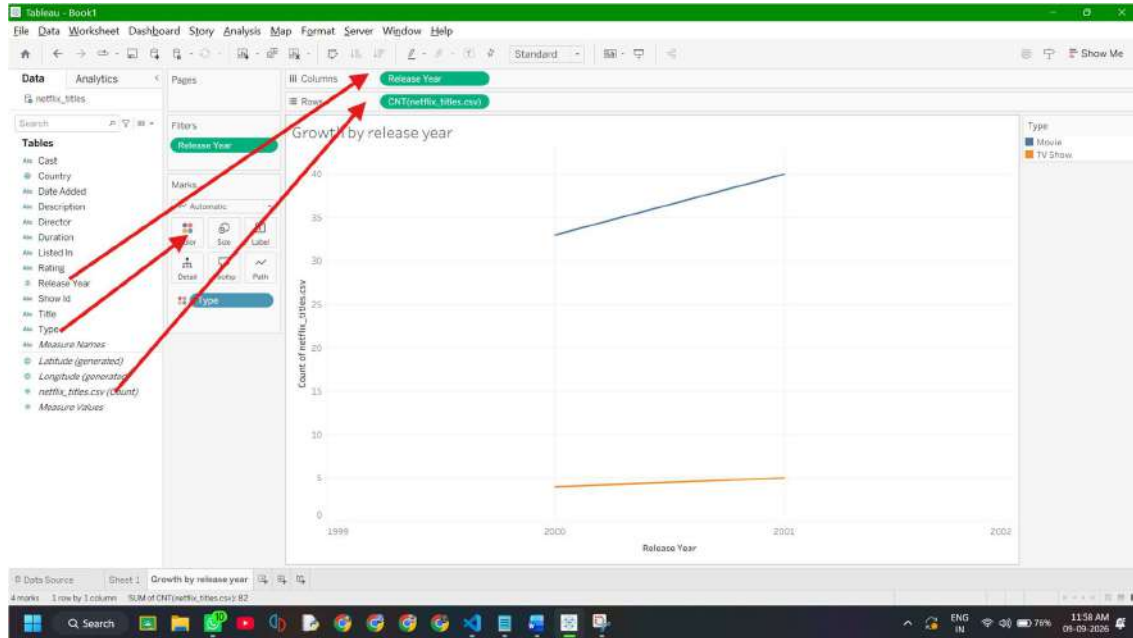
5. Result: A horizontal bar chart showing Movies vs. TV Shows.

Sheet 2: Release Year Growth (Line Chart)

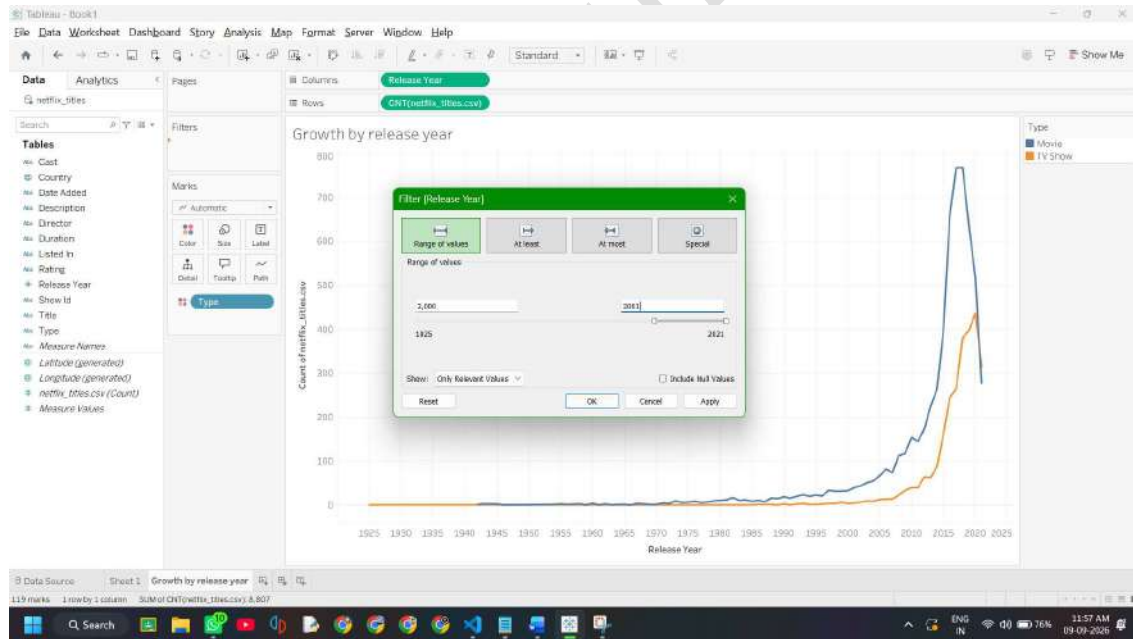
1. Create a worksheet named: **Growth by Release Year**.
2. Drag **release_year** to **Columns** (ensure it is Continuous or an unbroken timeline).
3. Drag **netflix_titles.csv (Count)** to **Rows**.

Data Visualization with Power BI and Tableau

4. Drag **type** to **Color**.



5. Filter: Drag **release_year** to **Filters** select **Range of values** set from 2000 to 2021 (removes sparse historical outliers).

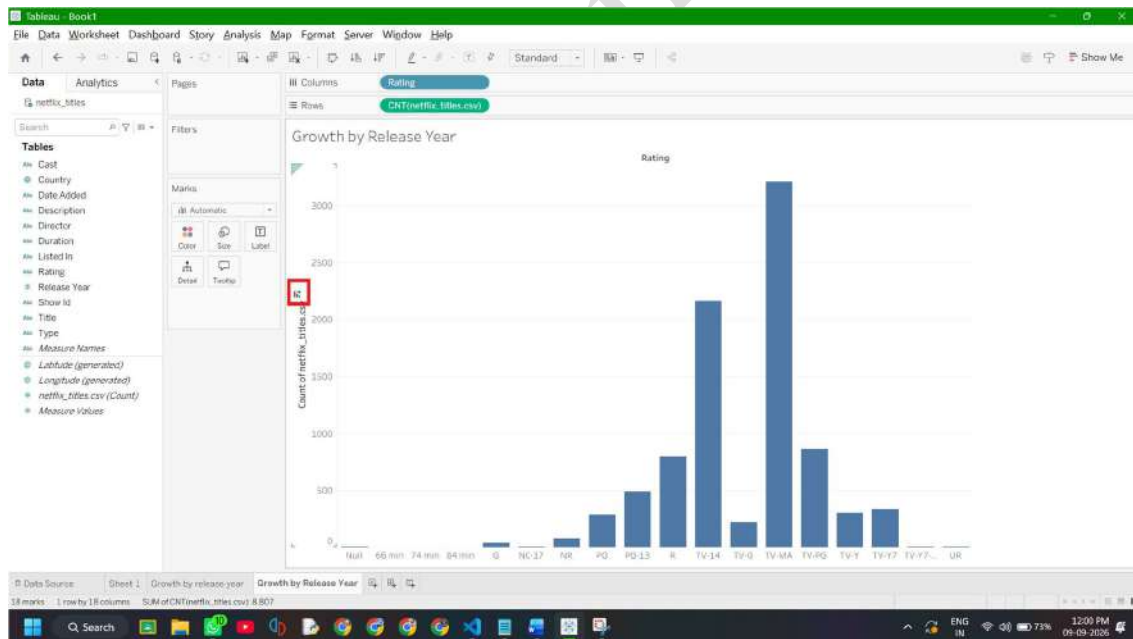
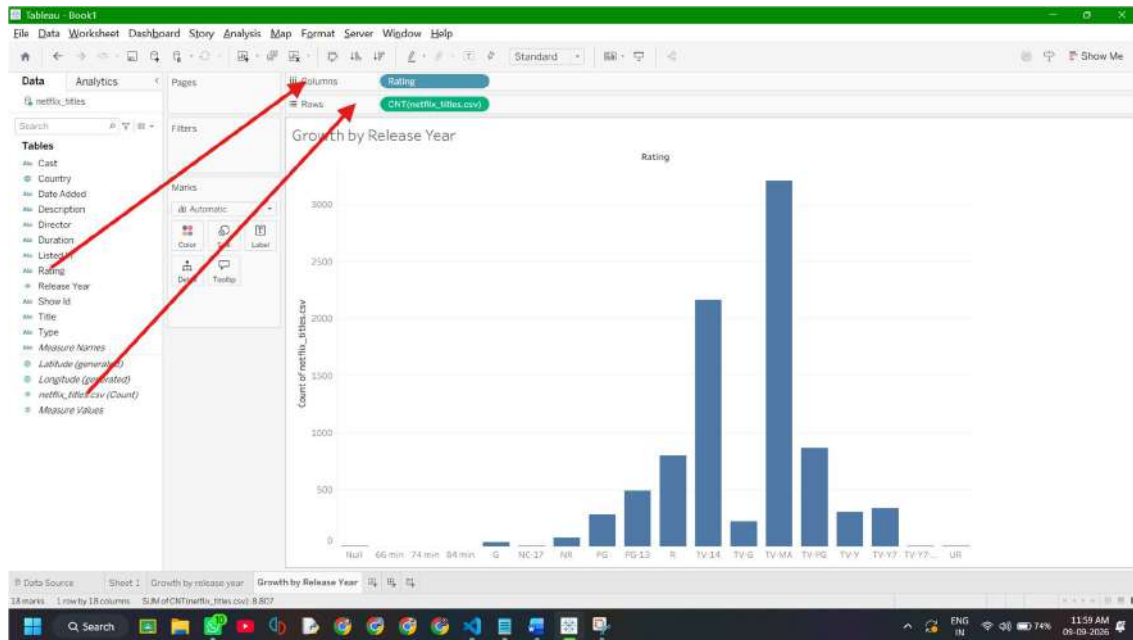


6. Result: A dual-line trend chart showing content volume over time.

Data Visualization with Power BI and Tableau

Sheet 3: Top 10 Ratings

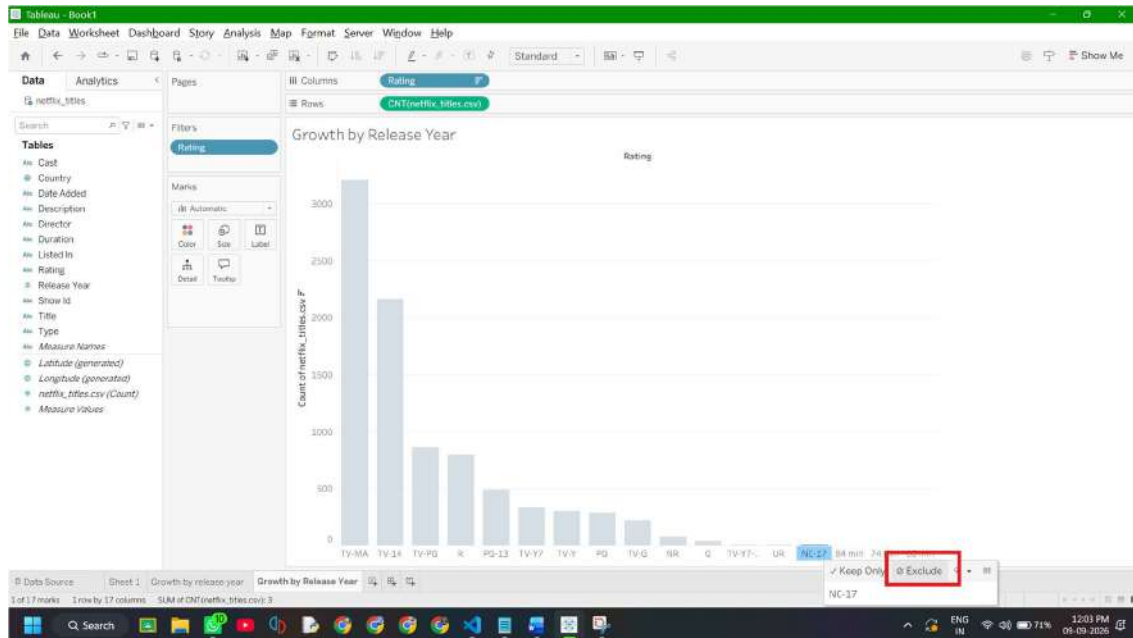
1. Create a worksheet named: **Growth by Release Year**.
2. Drag **rating** to **Rows**.
3. Drag **netflix_titles.csv (Count)** to **Columns**.



4. Sort descending by count (Click).

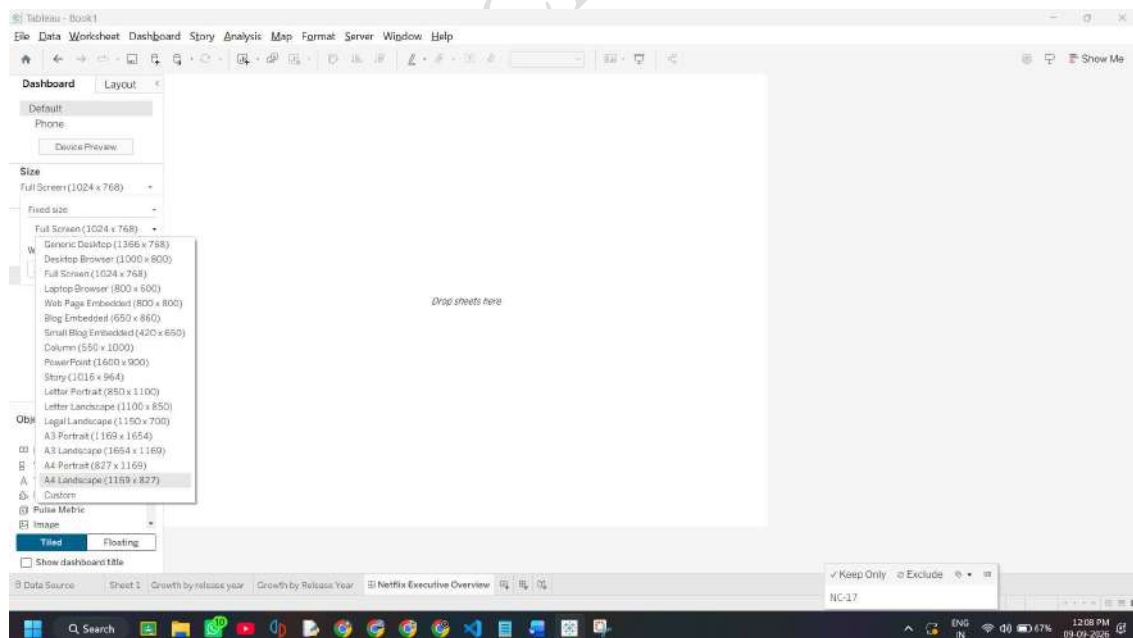
Data Visualization with Power BI and Tableau

5. Filter out Null ratings if any appear.



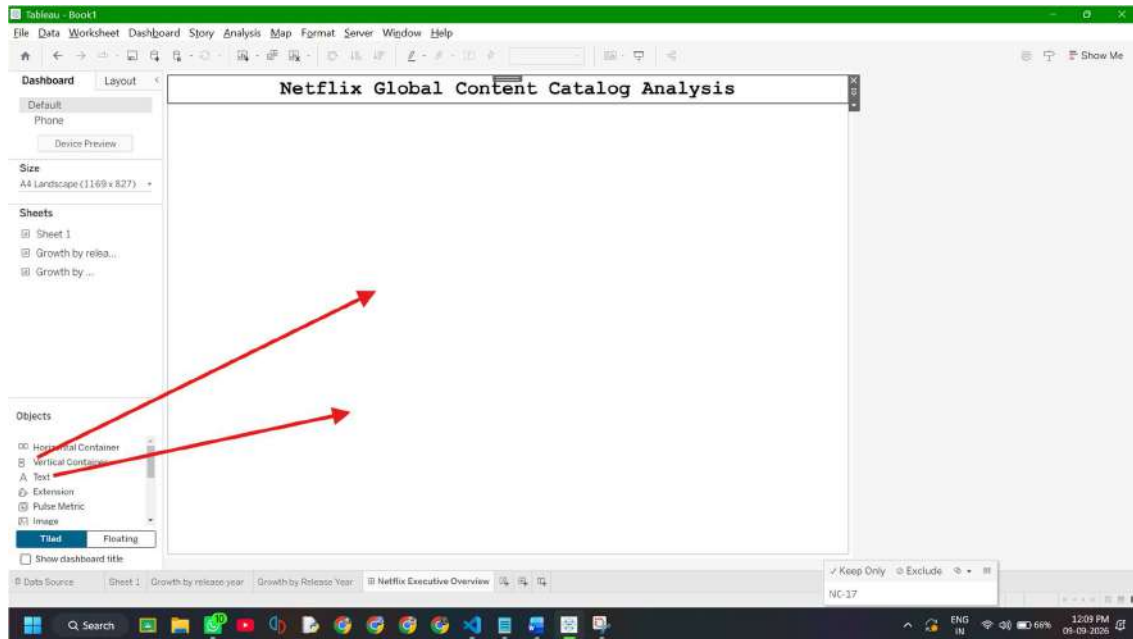
Step 2: Assemble the Executive Dashboard

1. Click the **New Dashboard** tab and rename it: **Netflix Executive Overview**.
2. In the **Dashboard** pane on the left, under **Size**, select the dropdown:
 - o Choose **Fixed Size** --> **A4 Landscape (1169 x 827 px)** or **Letter Landscape (1100 x 850 px)**. (Fixed size prevents layout shifts during PDF export).

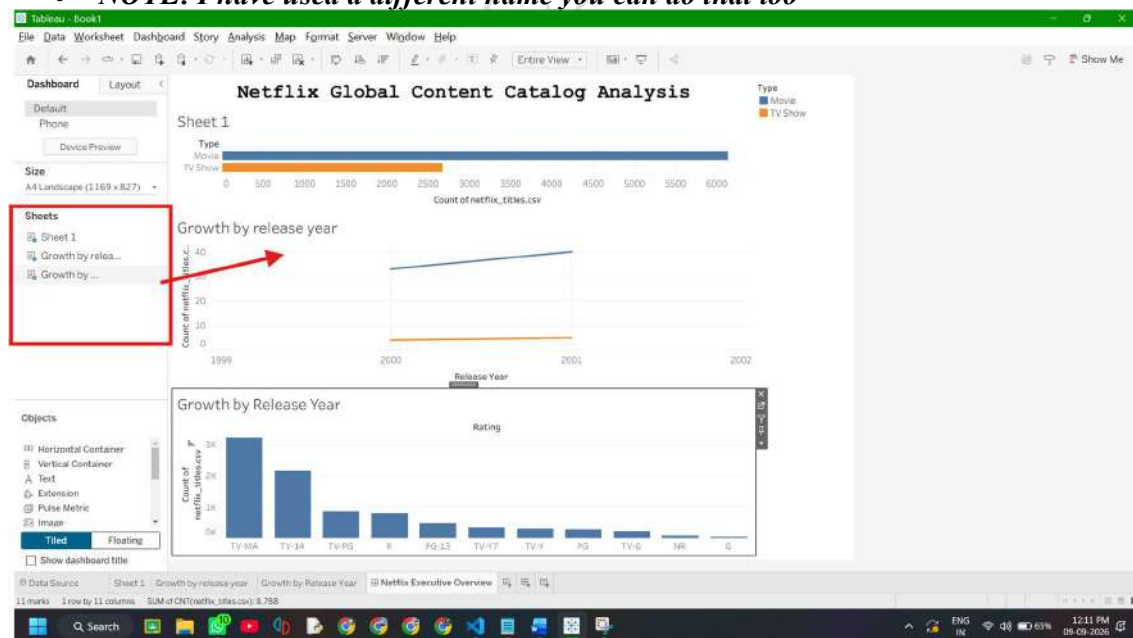


Data Visualization with Power BI and Tableau

3. Drag a **Horizontal** or **Vertical Container** onto the canvas.
4. Add a **Text** object at the top: Netflix Global Content Catalog Analysis(20pt Bold).



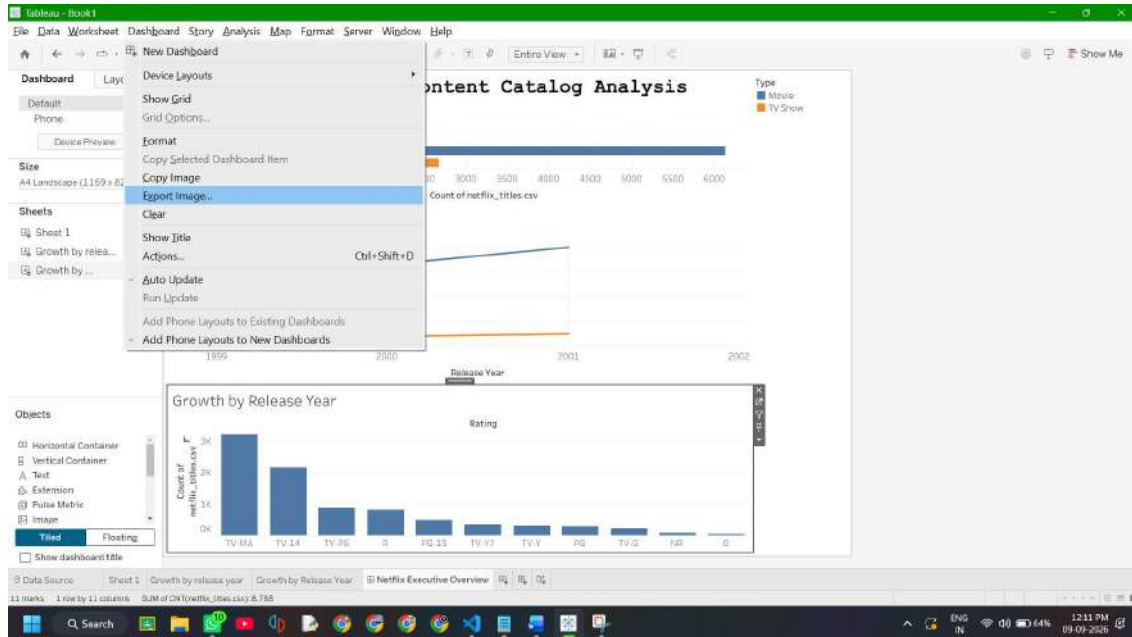
5. Drag your sheets into the canvas:
 - Place **Content Type Split** on the top left.
 - Place **Top 10 Ratings** on the top right.
 - Place **Growth by Release Year** across the bottom.
 - **NOTE: I have used a different name you can do that too**



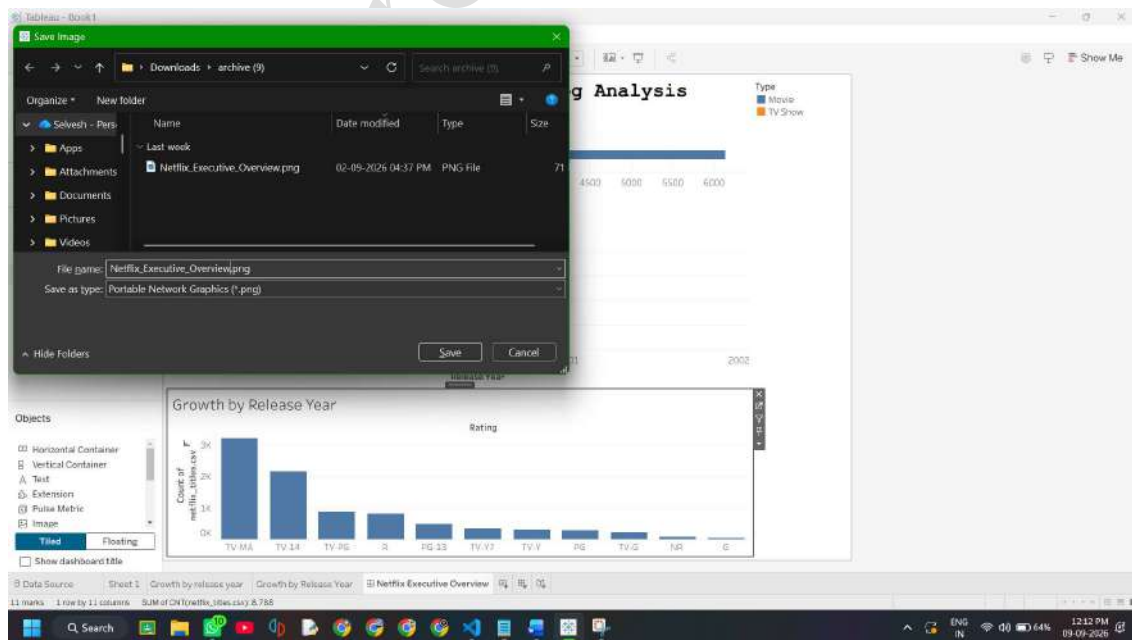
Data Visualization with Power BI and Tableau

Step 3: Export as High-Resolution Image (Raster / Vector)

1. Navigate to the top application menu bar:
 - o Select **Dashboard --> Export Image...** (or **Worksheet --> Export --> Image...**).



2. In the file save dialogue:
 - o Choose file location and name it: `Netflix_Executive_Overview.png`.
 - o Save as type: **PNG Image (*.png)** or **Enhanced Metafile (*.emf)** for vector scaling in PowerPoint.

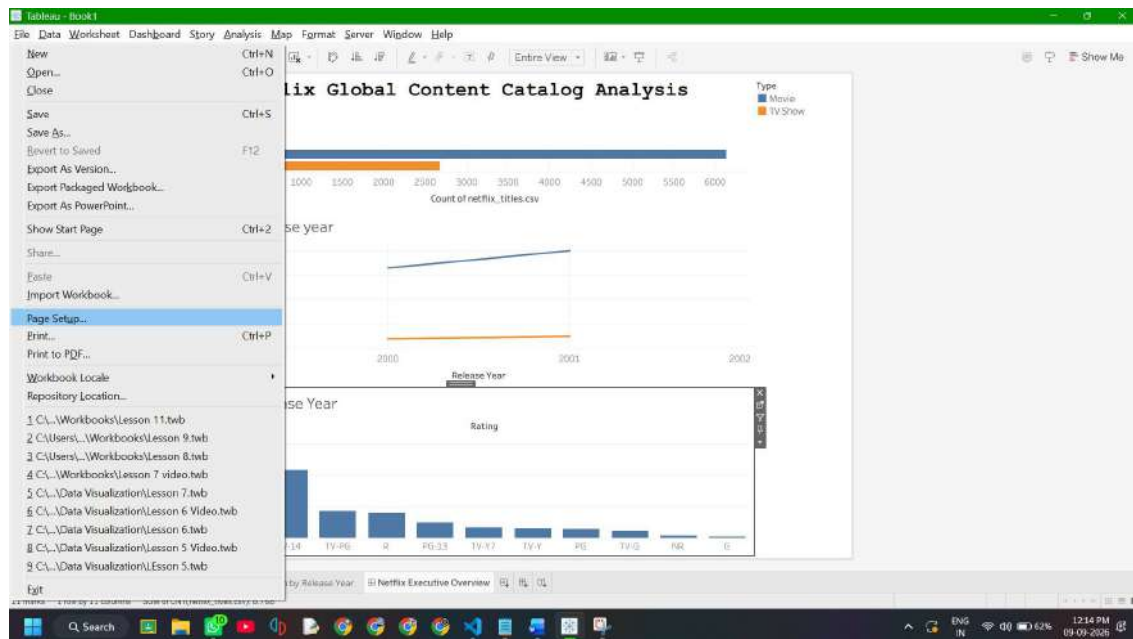


3. Click **Save**.

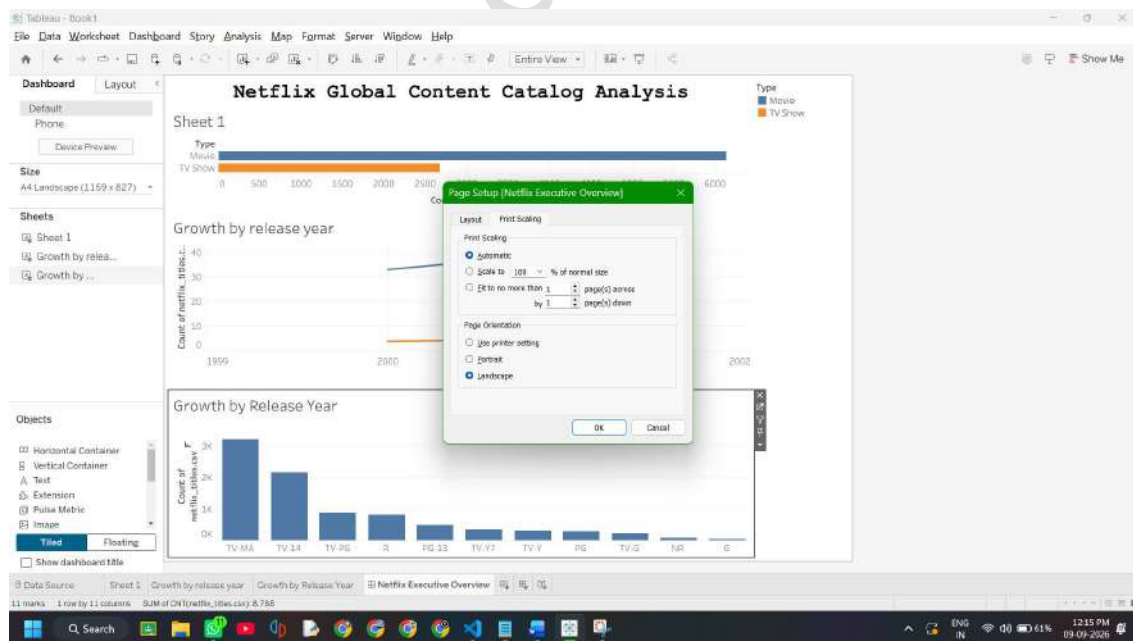
Data Visualization with Power BI and Tableau

Step 4: Configure Print Setup and Export to PDF

1. In the top menu, go to **File --> Page Setup...**:

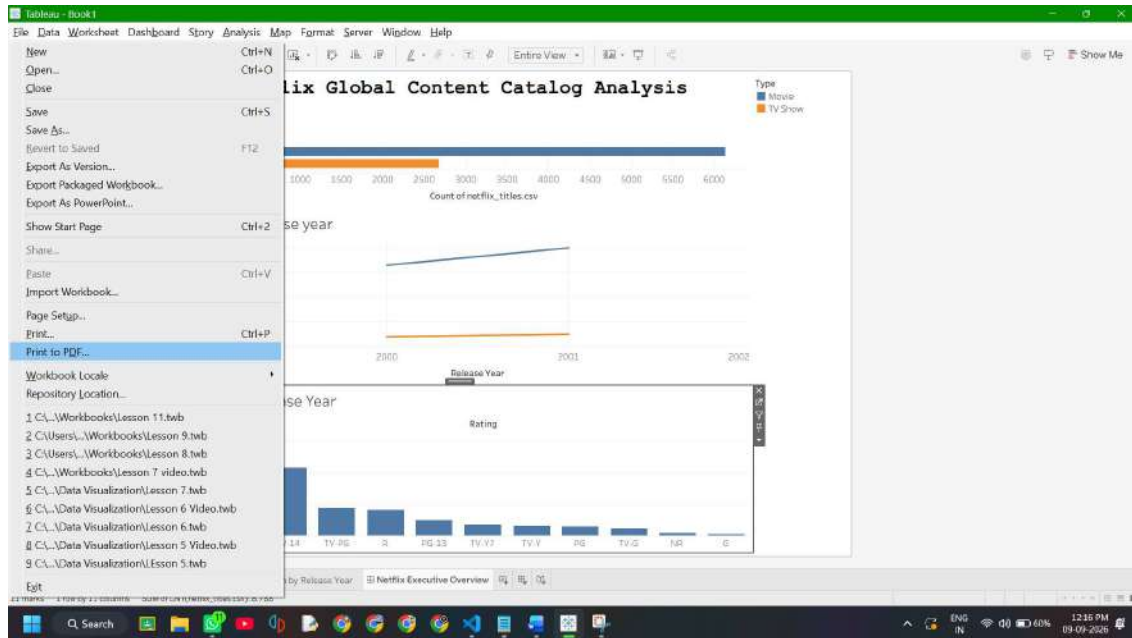


- **Page Type:** Select A4 (or Letter).
- **Orientation:** Select Landscape.
- **Scaling:** Select Fit to at most 1 page wide by 1 page high (ensures no cutoff).
- Click **OK**.

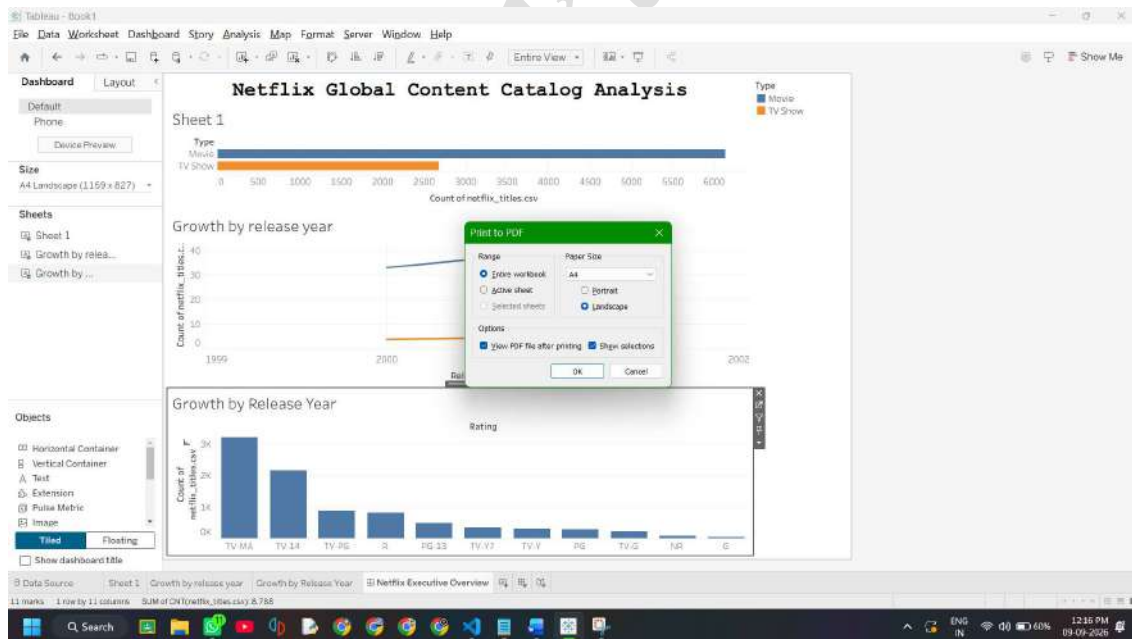


Data Visualization with Power BI and Tableau

2. Go to **File** --> **Print to PDF...**:



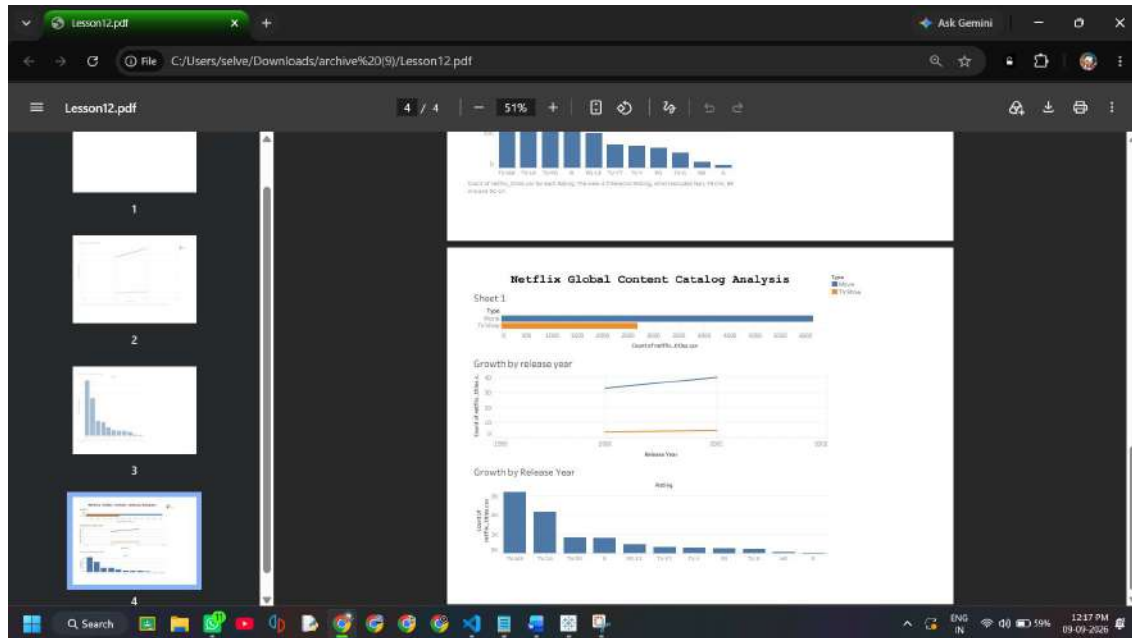
- **Range:** Select **This dashboard** (or *Entire workbook* to create a multi-page PDF report).
- **Paper Size:** Set to **Unspecified** (inherits Page Setup) or **A4**.
- **Orientation:** **Landscape**.



3. Click **OK**, choose the destination folder, and name the file **Netflix_Catalog_Report.pdf**.

Data Visualization with Power BI and Tableau

4. Open the generated PDF to confirm zero pixelation and complete margin alignment.



B: Use Presentation Mode in Tableau

Compatibility Note : You need Tableau Desktop Public Edition (100% Free Forever).

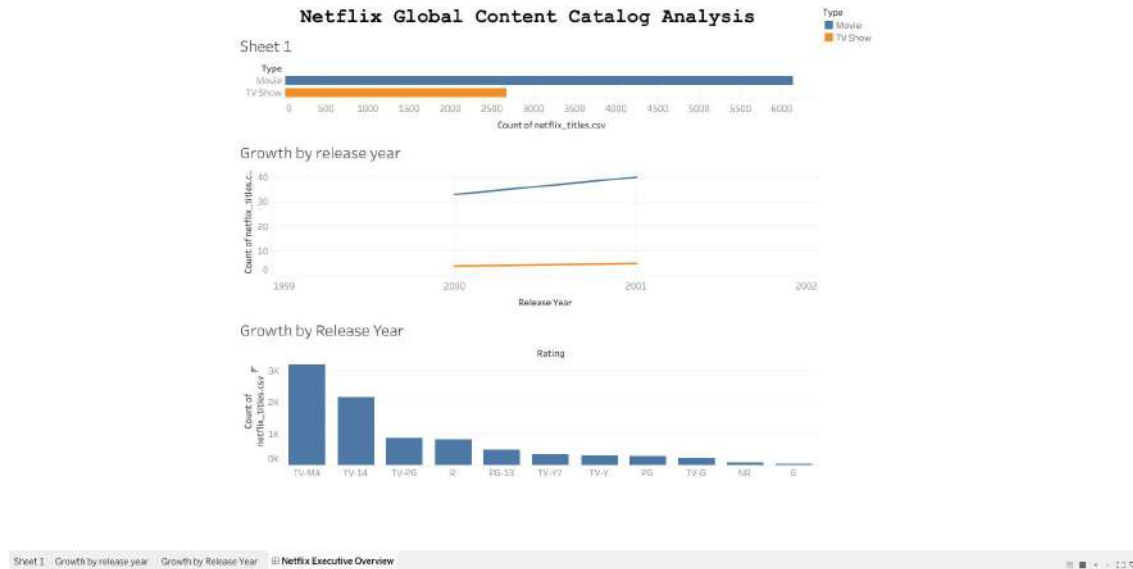
Objective: Use Tableau's built-in **Presentation Mode** to display visualizations in a distraction-free, full-screen environment for live meetings and stakeholder presentations.

During executive briefings, application sidebars, cards, shelves, and window chrome introduce unnecessary clutter. Presentation mode hides the development interface, retains all interactive tooltips and cross-filtering, and provides a lightweight presentation toolbar for smooth navigation.

Step 1: Launch Presentation Mode

1. Open your **Netflix Executive Overview** dashboard in Tableau Desktop.
2. Enter Presentation Mode using any of these methods:
 - **Keyboard Shortcut:** Press **F7** (Windows) or **Cmd + Enter** (Mac).
 - **Toolbar Icon:** Click the **Presentation Mode icon** (a small projection screen icon) in the top application toolbar.
 - **Menu Path:** Go to the top menu and select **Window --> Presentation Mode**.

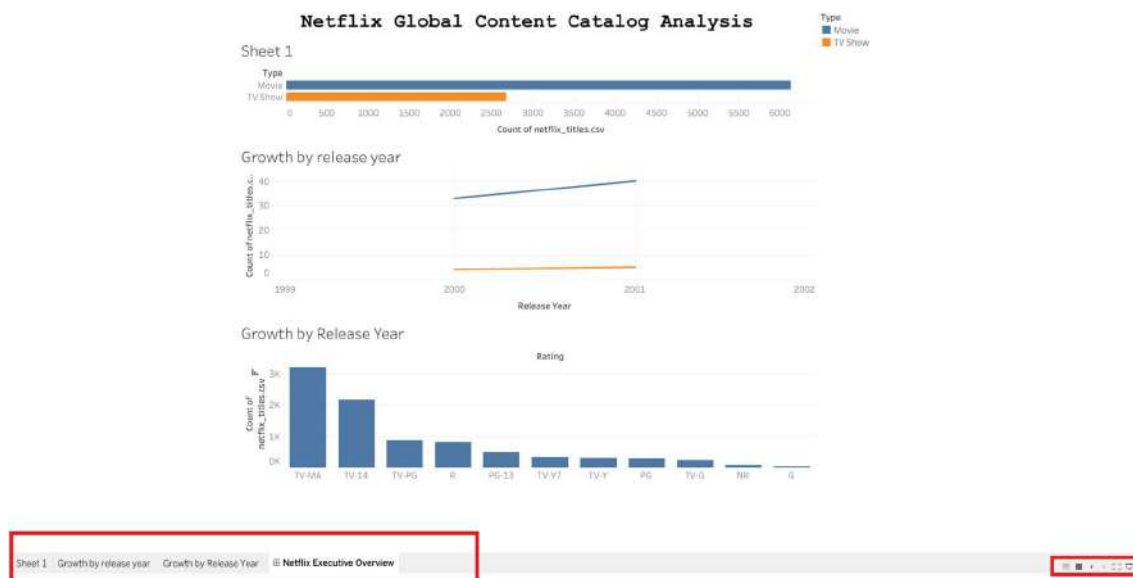
Data Visualization with Power BI and Tableau



Step 2: Use On-Screen Presentation Controls

Once in full-screen mode, an unobtrusive toolbar appears at the bottom or top corner of the display:

- **Sheet Navigation:** Use the left/right arrows on the toolbar (or the **Page Up** / **Page Down** keys) to cycle through all worksheets, dashboards, and stories in your workbook.
- **Live Interactive Filtering:** Click on any bar or line mark (e.g., clicking *Movies* in the split chart) to demonstrate real-time cross-filtering.
- **Interactive Tooltips:** Hover over data points to present granular numbers on demand without cluttering the main slide.
- **Reset / Revert:** Click the **Revert** arrow on the presentation toolbar to reset all filters back to their initial state.

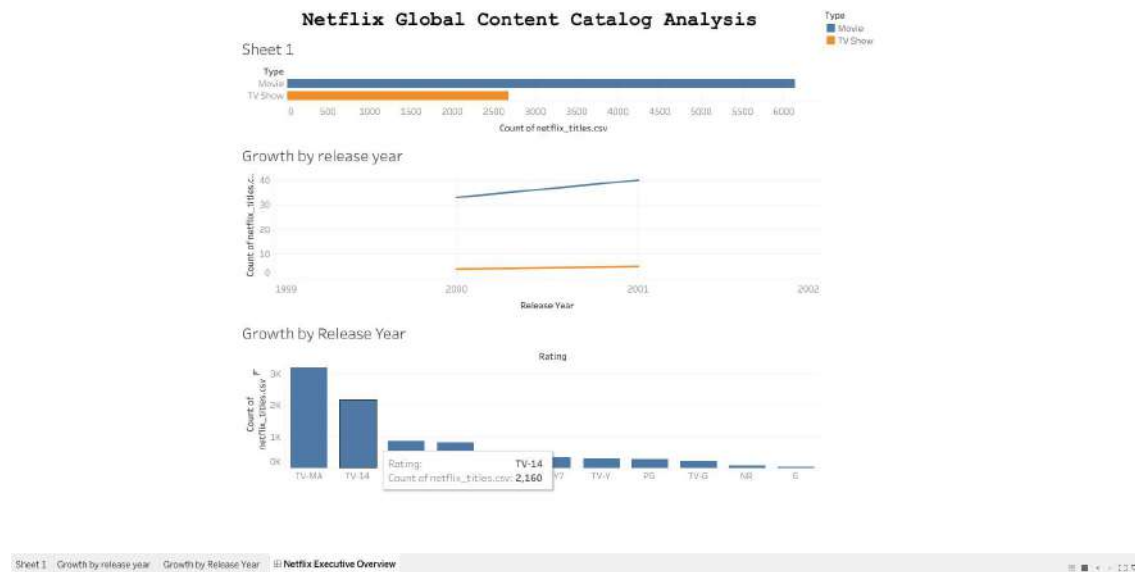


Step 2: Live Navigation and Interactivity During Presentation

While in Presentation Mode, test the following live controls:

1. Hover Exploration:

- Move your mouse over individual data marks (e.g., specific bars in Content Type Split or peak data points in Growth by Release Year).
- Confirm that tooltips render cleanly with exact counts and dates.



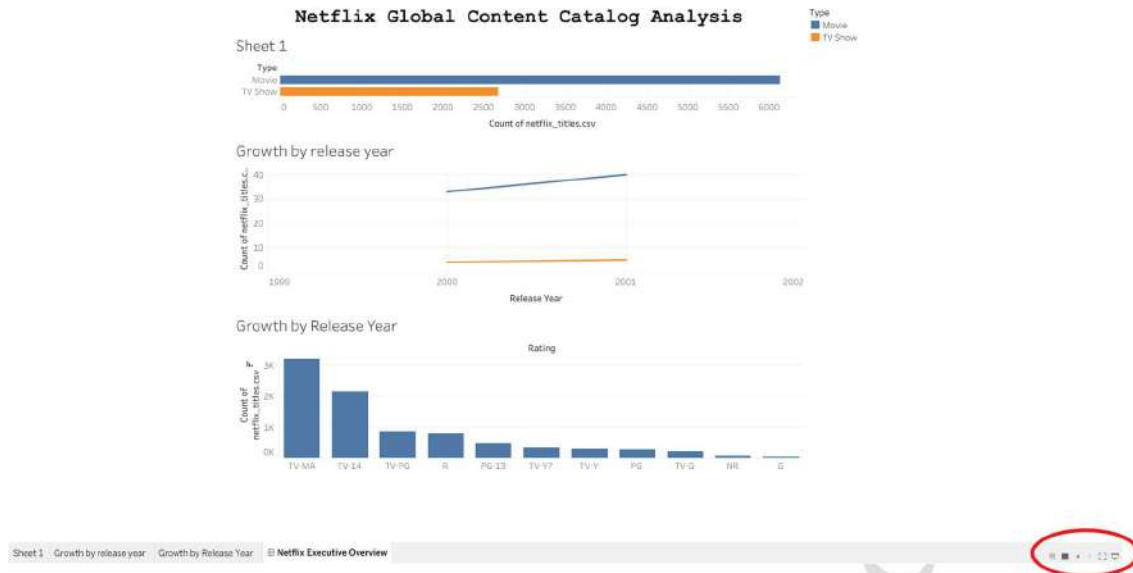
2. Interactive Filtering:

- If quick filters or legends are placed on the dashboard, click to filter or highlight categories live.

3. Multi-Sheet Navigation:

- Move your mouse to the bottom-left corner to reveal the subtle navigation toolbar.
- Use the **Next Sheet (→)** and **Previous Sheet (←)** arrows (or the **Page Up / Page Down** keys) to cycle between individual worksheets and dashboards seamlessly.

Data Visualization with Power BI and Tableau



Step 3: Exit Presentation Mode

- Press the **Esc** key on your keyboard (or click the **Exit Presentation Mode** icon in the bottom-left toolbar) to return to standard authoring mode.

C: Publish Dashboards to Tableau Public

Compatibility Note: You need to use "Tableau Desktop Public Edition to implement this task.

Local Environment Strategy: When completing this task in a local environment without active server credentials or cloud sync, the standard industry and academic deliverable for Tableau Public is a **Tableau Packaged Workbook (.twbx)** pre-configured with extracted data, metadata properties, and view presets ready for immediate web deployment or local grading.

Objective: Replace the cloud-dependent "Publish to Tableau Public" workflow by compiling your visualizations, calculations, formatting metadata, and data extract into a single, offline **Tableau Packaged Workbook (.twbx)**

A standard Tableau workbook (.twb) contains only XML layout instructions and points to the external file path of `netflix_titles.csv`. If you move that CSV or share the .twb with an instructor or peer, the connection breaks. Generating a .twbx archives the underlying data extract directly inside the file container, enabling fully interactive, offline distribution without web accounts or server installations.

Step 1: Confirm Data Extract Status

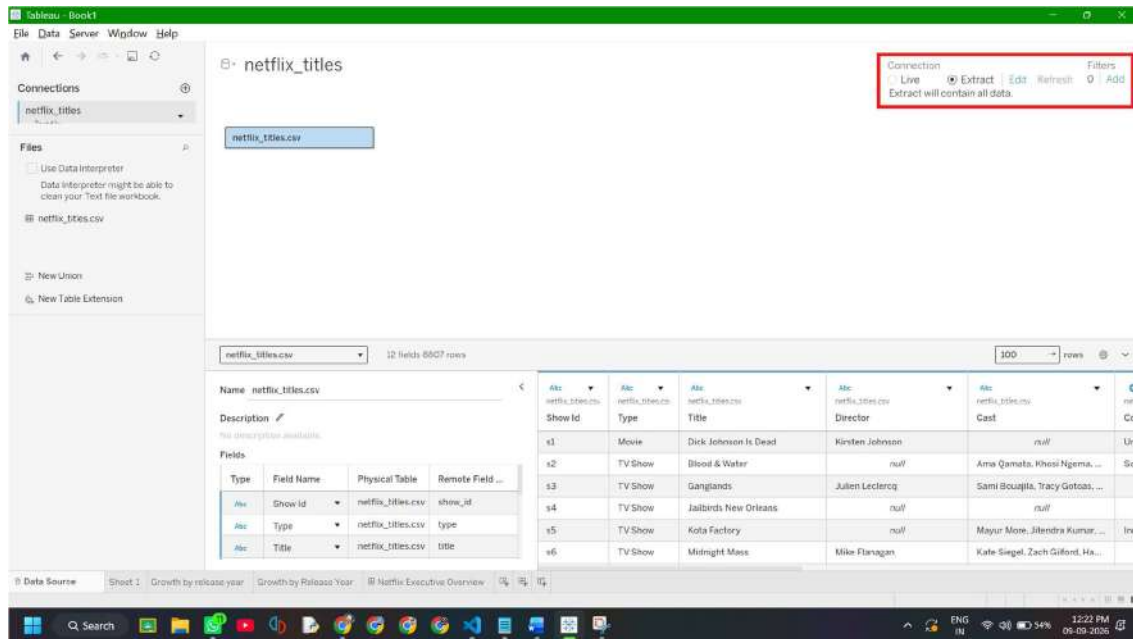
A packaged workbook works best when the underlying data is extracted rather than referenced as a live text file.

1. Check the top-left **Data Pane** where `netflix_titles` is listed:
 - If you see **two linked cylinders** (■ ■), the data is already extracted. Proceed to Step 2.

Data Visualization with Power BI and Tableau

2. If you see a **single cylinder (■)**:

- Go to the **Data Source** tab in the bottom-left corner.
- In the top-right corner, select the **Extract** radio button.



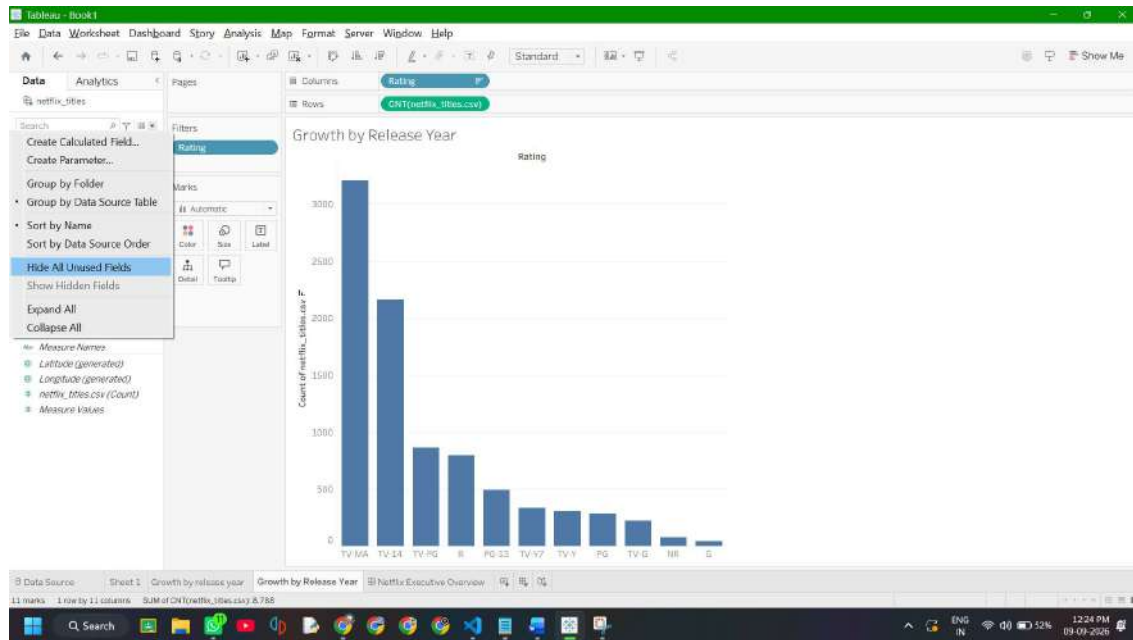
- Click back onto your **Netflix Executive Overview** dashboard tab.

Step 2: Prune Unused Columns to Minimize File Footprint

To keep the packaged archive compact:

1. In the **Data Pane**, click the drop-down menu icon (small down-arrow next to the search bar).
2. Select **Hide All Unused Fields**.
 - *Outcome:* Columns not used in your visualizations or calculations (e.g., full text descriptions or raw cast lists, if unused) will be excluded from the package, reducing disk space and load times.

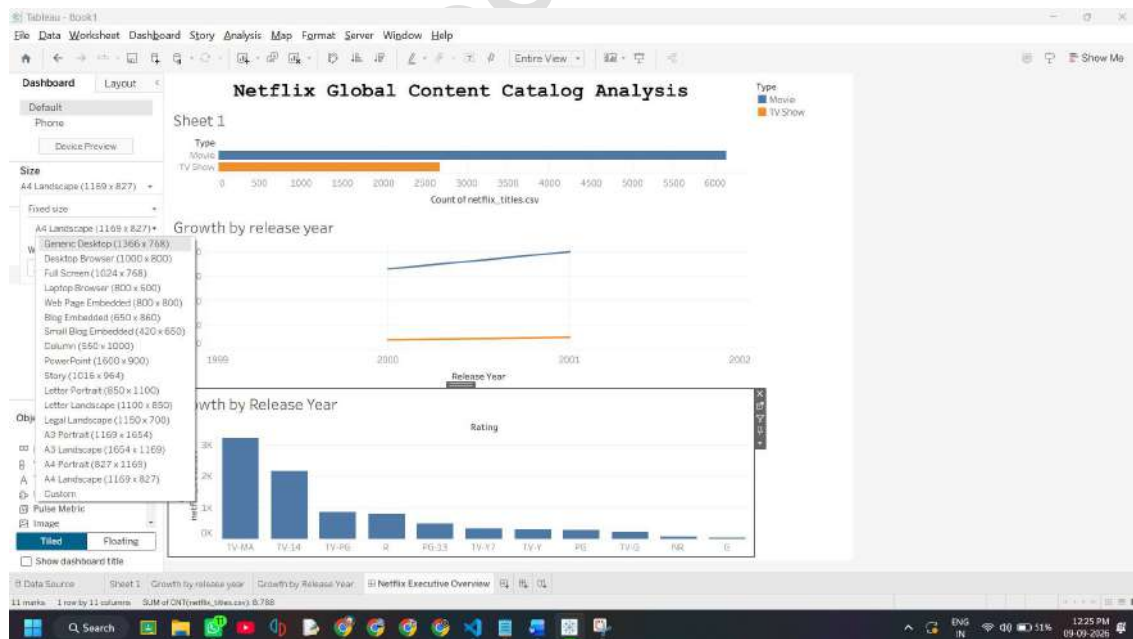
Data Visualization with Power BI and Tableau



Step 3: Pin the Default Landing Tab

When someone opens your packaged file, Tableau opens whatever tab was active when the file was saved:

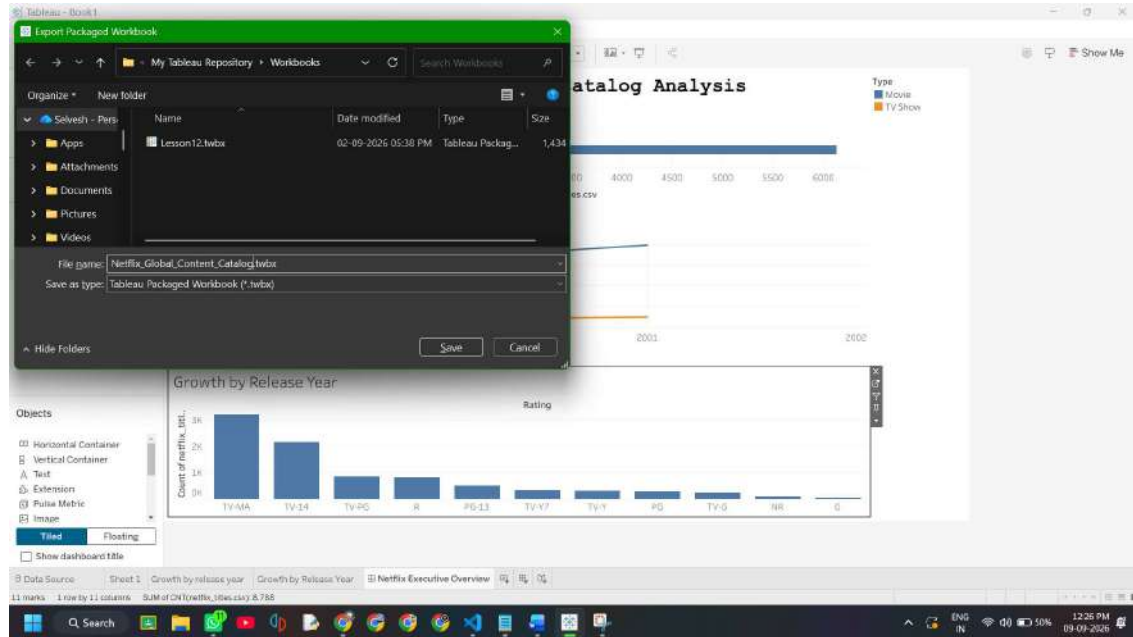
1. Click on the **Netflix Executive Overview** tab at the bottom so it is active.
 2. In the left-hand **Dashboard** panel, ensure **Size** is set to **Fixed Size** (e.g., 1366 x 768 px or A4 Landscape).
- Why: Fixed sizing prevents web browsers on different screen resolutions from distorting the chart layout or causing unwanted scrollbars.



Step 4: Export the Standalone Packaged Workbook (.twbx)

1. Go to the top menu bar and click **File** --> select **Export Packaged Workbook...** (or **File** --> **Save As...**).

Data Visualization with Power BI and Tableau



2. In the file save window:
 - **File name:** Netflix_Global_Content_Catalog.twbx
 - **Save as type:** Select **Tableau Packaged Workbook (*.twbx)**
3. Click **Save**.

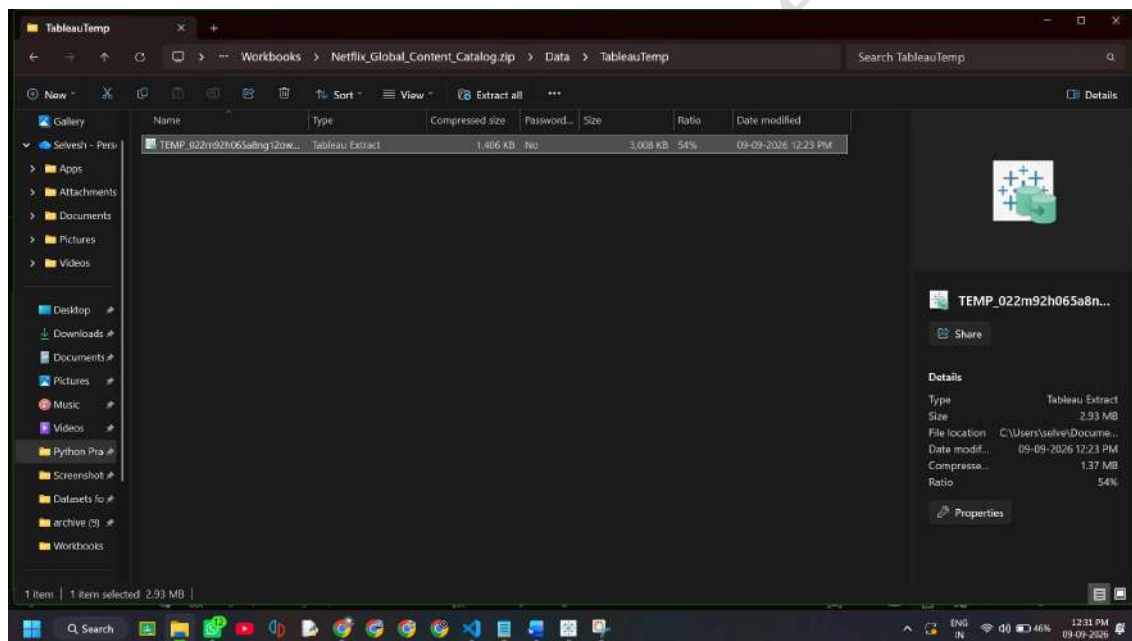
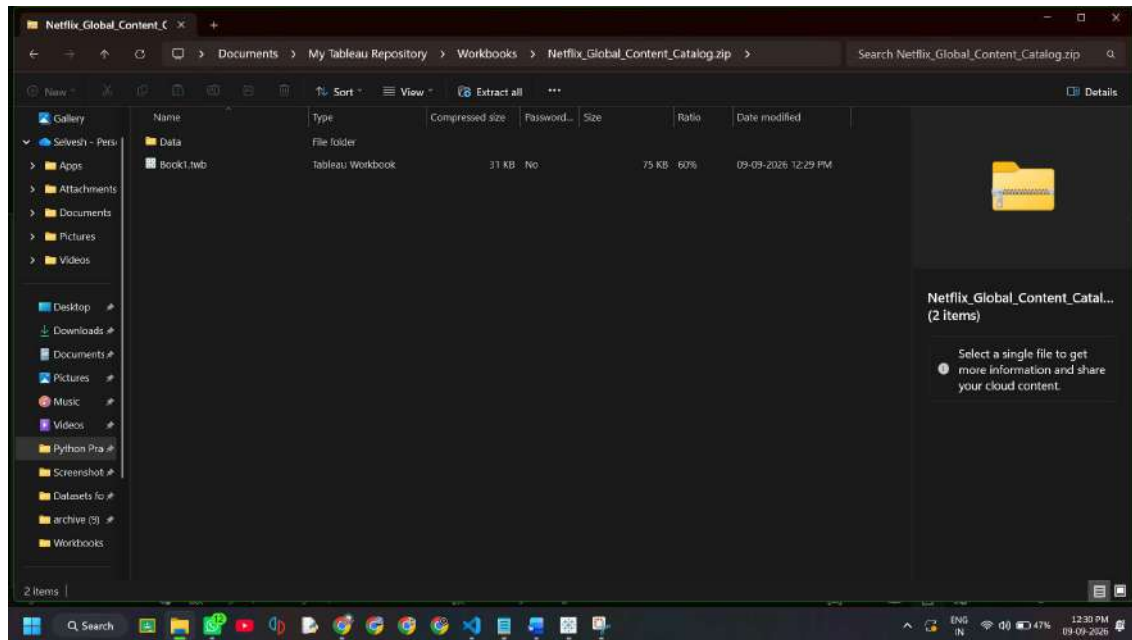
What this file contains: Both the .hyper data extract and the complete visual design (XML code, filters, formatting), making it completely independent of your local netflix_titles.csv file.

Step 5: Verify Archive Integrity Locally

To confirm that the package is completely self-contained:

1. In your operating system file explorer (Windows Explorer), locate Netflix_Global_Content_Catalog.twbx.
2. Notice the file icon: a standard Tableau workbook icon with a small orange package emblem.
3. *(Technical Verification)* A .twbx file is internally a standard ZIP archive. If you rename the extension from .twbx to .zip and open it, you will find:
 - The .twb (XML schema and visualization code).
 - A Data folder containing the compressed .hyper extract.

Data Visualization with Power BI and Tableau

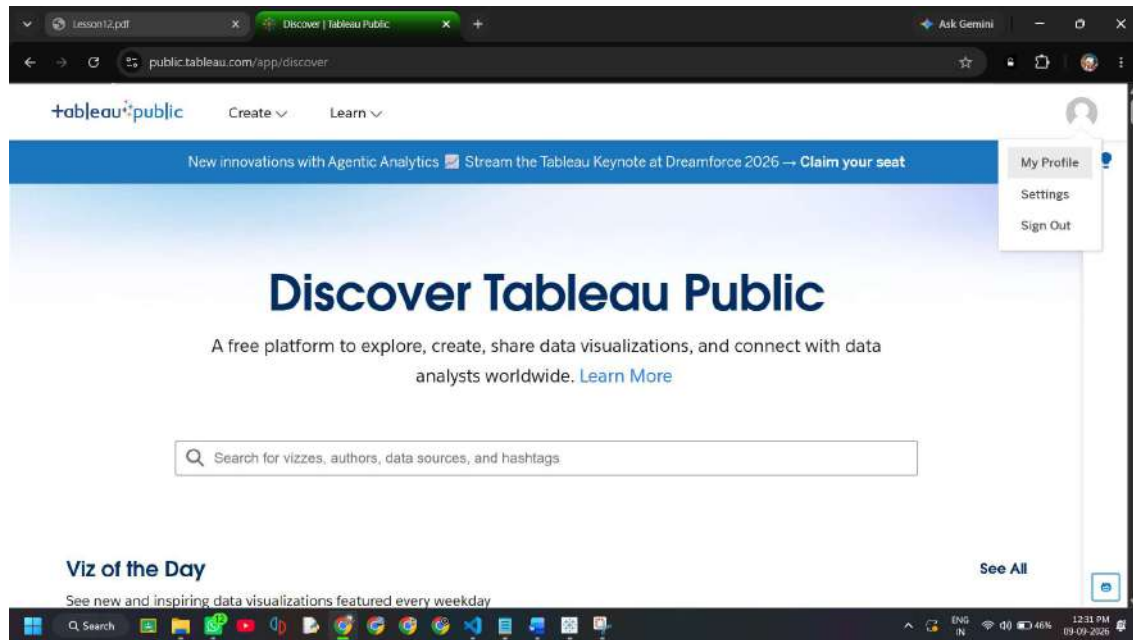


4. This confirms the workbook can be moved to any computer or shared via USB/email without needing `netflix_titles.csv`.

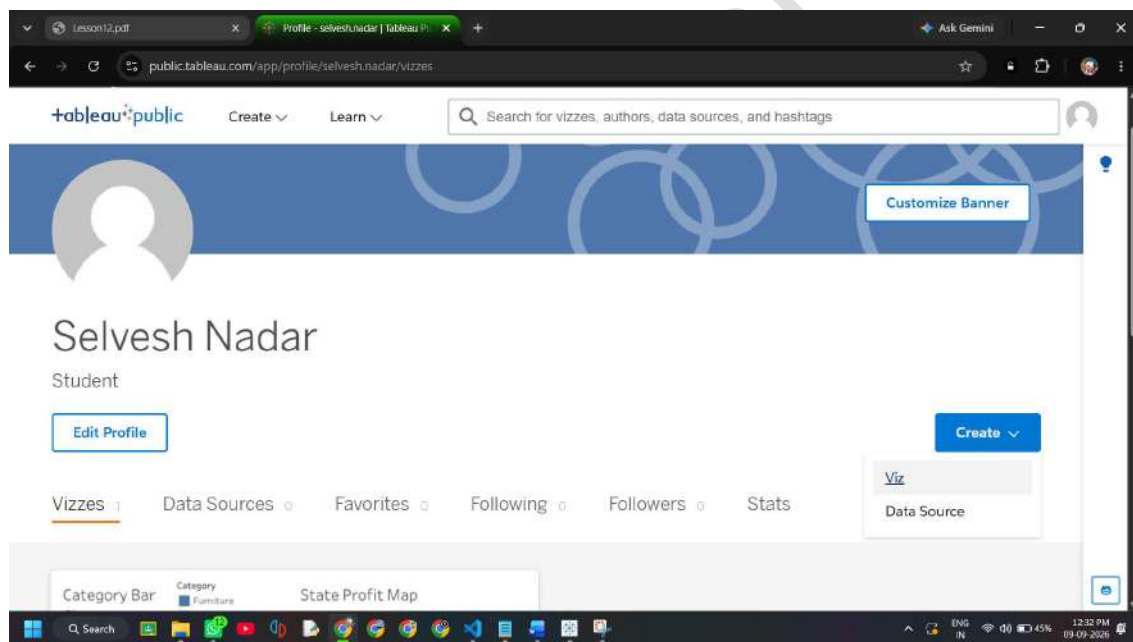
Additional Step : Publish to Tableau Public via Web Interface[if you have tableau public login]

1. Open your web browser and go to public.tableau.com.
2. Sign in to your free Tableau Public account (or click *Sign Up* to create one).
3. In the top navigation bar, click your **Profile Avatar** (top-right) --> select **My Profile**.

Data Visualization with Power BI and Tableau

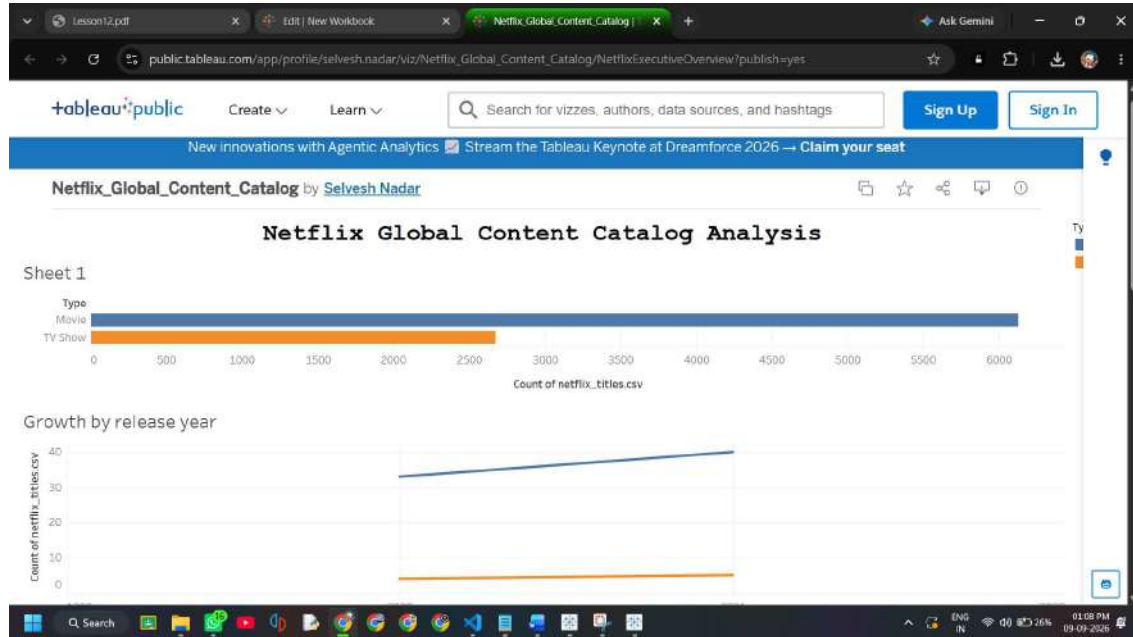


4. Click the **Create / Upload** button (or the **Upload a Viz** option).



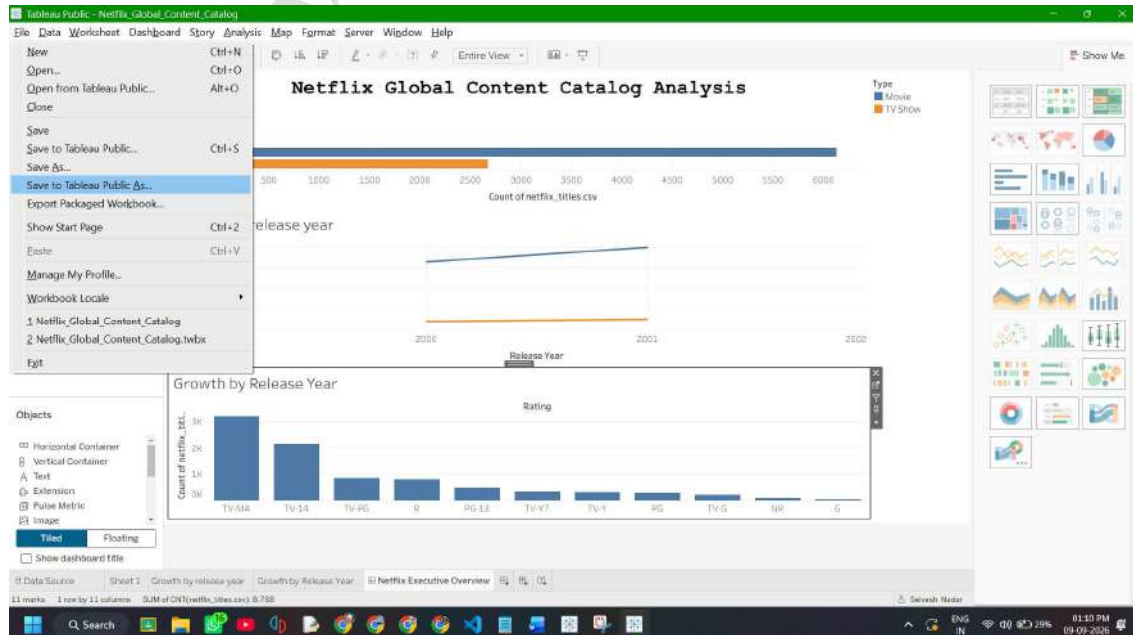
5. Drag and drop your saved `Netflix_Global_Content_Catalog.twbx` file into the upload area.
6. Once the upload finishes, your interactive dashboard is live on Tableau Public.
7. On the published web page:

Data Visualization with Power BI and Tableau



- **Test Interactivity:** Click on the *Movie* or *TV Show* bars to confirm web cross-filtering works.
- **Retrieve Links:** Click the **Share** button below the dashboard to copy:
 - **Direct URL:** For resumes, LinkedIn, or portfolio submissions.
 - **LINK:** https://public.tableau.com/views/Netflix_Global_Content_Catalog/NetflixExecutiveOverview?:language=en-US&:sid=&:redirect=auth&:display=count=n&:origin=viz_share_link
 - **Embed Code:** For embedding into HTML websites or blogs.

NOTE: Save the twbx file and open it in Tableau Public after opening it navigate to file and click "Save to Tableau Public As" after clicking it sign in your tableau acc in small popup window then you will be directed to webpage of Tableau where you can see your name and the dashboard you have uploaded.



Data Visualization with Power BI and Tableau

D: Share Dashboards via Tableau Server / Tableau Cloud

Objective: Implement the local offline equivalent of enterprise sharing by configuring sheet visibility, distribution governance, and simulating end-user consumption using free distribution tools like **Tableau Reader**.

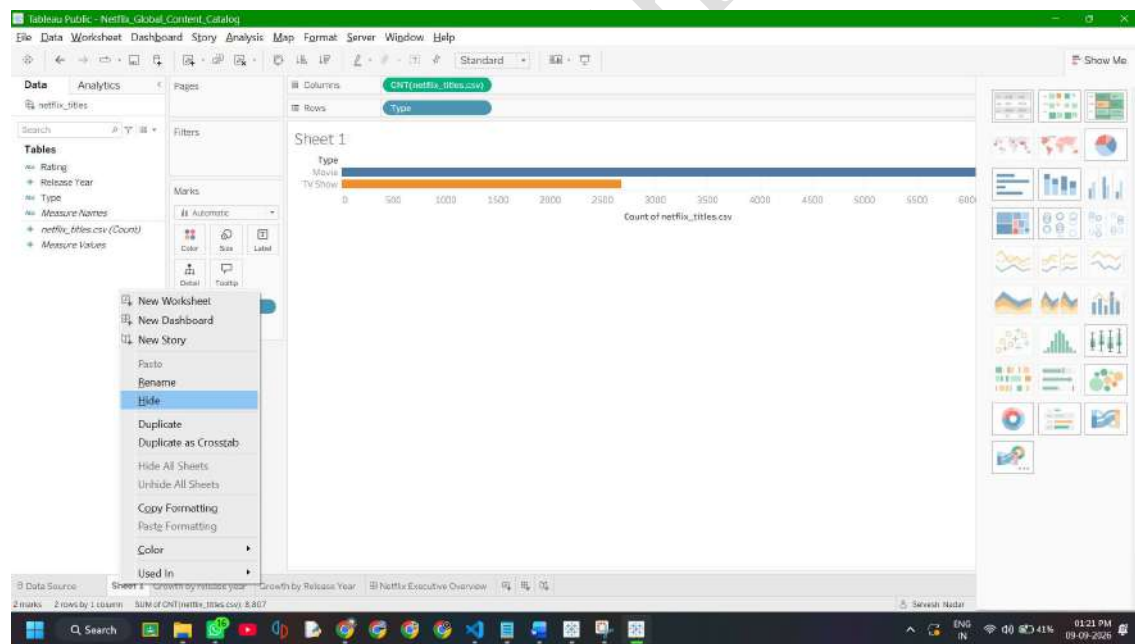
In corporate settings without centralized Tableau Server or Cloud infrastructure, workbooks must still be shared securely and cleanly. This requires preventing end-users from seeing messy draft sheets, securing raw underlying data from casual alteration, and verifying that the deliverable functions properly for users who do not own an authoring license.

Step 1: Clean Up Workbook Tabs (Sheet Governance)

In enterprise deployments, raw working sheets should never be visible to business stakeholders. Only finalized executive dashboards should appear in the navigation bar.

1. Hide each underlying worksheet:

- Right-click the **Content Type Split** tab --> select **Hide Sheet**.
- Right-click **Growth by Release Year** --> select **Hide Sheet**.
- Right-click **Top 10 Ratings** --> select **Hide Sheet**.
- (Note: If "Hide Sheet" is grayed out, ensure the sheet is currently placed inside your *Netflix Executive Overview dashboard*).



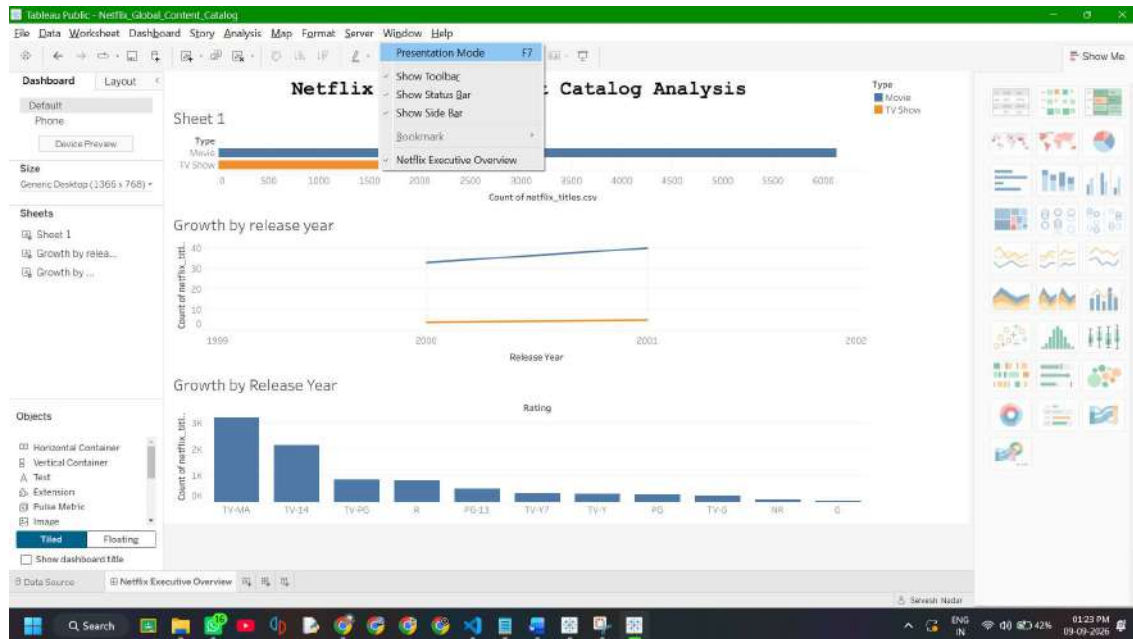
- **Verify the Layout:** Look at the bottom tab bar. Only the **Netflix Executive Overview** dashboard tab should now be visible.

Data Visualization with Power BI and Tableau

Step 2: Configure Navigation and Sheet Tabs Settings

When stakeholders open the file, configure how Tableau displays navigation:

1. Go to the top application menu bar and click **Window**.
2. Look at the **Show Sheet Tabs** toggle:
 - If you have only one executive dashboard, you can leave it unchecked or keep it clean so the dashboard spans the entire interface.
3. Verify that the window controls are clean and unobstructed by unused toolbars.



Step 3: Understand the Tableau Reader Distribution Mechanism

When Tableau Server or Tableau Cloud is unavailable, organizations distribute dashboards via **Tableau Reader**:

- **What is Tableau Reader?**
 - A free, standalone desktop viewer application for Windows and macOS.
 - Downloadable at no cost by any stakeholder or client without a license key.
- **Consumer Capabilities in Tableau Reader:**
 - **Interactivity:** Viewers can click marks, filter categories, hover for tooltips, and change parameters.
 - **Zero Accidental Edits:** Viewers **cannot** move charts, alter formulas, break calculations, or modify visual aesthetics.
 - **File Requirement:** Tableau Reader opens **only** .twbx (packaged) workbooks with extracted data, ensuring stakeholders never experience "data source not found" errors.

Step 4: Simulate Consumption via Tableau Reader / Fresh Session

To verify that your workbook functions as a true enterprise-ready deliverable:

1. In Tableau Desktop, press **Ctrl + S** (or go to **File** → **Save**) to save your packaged workbook (Netflix_Global_Content_Catalog.twbx).

Data Visualization with Power BI and Tableau

2. Close your current Tableau Desktop window.
3. Double-click the saved **Netflix_Global_Content_Catalog.twbx** file in your local file explorer to reopen it.
4. **Audit the File:**
 - Does it open directly to the **Netflix Executive Overview**?
 - Are the underlying calculation/raw sheets hidden from view?
 - Does it open instantly without asking for database passwords or CSV file paths?
5. If yes to all, your deliverable is structured correctly for offline distribution.

Follow these steps if you have paid Tableau Creator license

Version Compatibility Note: Publishing directly to an enterprise Tableau Server or Tableau Cloud environment requires a paid **Tableau Creator** license

Objective: Understand how enterprise workbooks, embedded data sources, and automated scheduled extract refreshes are deployed to centralized Tableau Server or Tableau Cloud sites.

In production business intelligence environments, analytics assets cannot rely on local email attachments. Tableau Server and Cloud act as centralized, governed web portals that enable secure access, scheduled nightly database refreshes, mobile access via the Tableau Mobile app, and web-based dashboard editing without local software installations.

Step 1: Authenticate to Tableau Server / Cloud

1. In the top application menu bar of Tableau Desktop, go to **Server --> Sign In...**
2. In the connection window:
 - Select **Tableau Cloud** (or enter your on-premises server URL, e.g., [https://analytics.enterprise.com] (https://analytics.enterprise.com)).
 - Click **Connect**.
3. Enter your organizational single sign-on (SSO) credentials or username/password and confirm authentication.

Step 2: Initiate the Workbook Publishing Wizard

1. From the top menu, go to **Server --> Publish Workbook...** (or press the publish shortcut).
2. The **Publish Workbook to Tableau Server** dialog opens with several configuration settings:
 - **Project:** Select the target destination directory (e.g., Default, Executive Analytics, or Marketing Sandbox).
 - **Name:** Specify the published title (e.g., Netflix_Global_Catalog_v1).
 - **Description & Tags:** Add enterprise metadata tags (e.g., Production, Catalog, Q3-Report).

Step 3: Configure Sheets, Permissions, and Data Sources

Inside the publishing wizard dialog:

1. **Sheets to Publish:**
 - Click **Edit** under *Sheets*.

Data Visualization with Power BI and Tableau

- Select **Only Dashboards** (e.g., check *Netflix Executive Overview* and uncheck individual raw worksheet tabs like *Content Type Split* to prevent clutter for end users).
- Check **Show sheets as tabs** if viewers need to navigate across multiple dashboard pages.
- 2. **Data Sources & Authentication:**
 - Click **Edit** under *Data Sources*.
 - Set authentication to **Embedded password** (for live SQL databases) or **Refreshable Extract**.
- 3. **Interactive Permissions:**
 - Choose whether to inherit permissions from the parent project folder or define custom group access.

Step 4: Execute Publishing and Configure Refresh Schedules

1. Click the blue **Publish** button in the lower-right corner.
 2. Once the upload finishes, the server web portal launches the live interactive dashboard.
 3. **Configure Automated Refreshes on Server Web UI:**
 - On the published workbook page in your browser, navigate to the **Extract Refreshes** tab.
 - Click **New Extract Refresh**.
 - Choose a schedule frequency (e.g., **Daily at 06:00 AM** or **Weekly on Mondays**).
 - Click **Create Refresh**.
 - *Outcome:* Tableau Server now runs background jobs automatically updating the extract without needing any manual file open/save actions.
-

E: Enable User Interaction on Shared Dashboards

Compatibility Note (Tableau Desktop 2026.2): Fully supported natively in all editions (Free, Trial, and Professional).

Objective: Configure interactive features including **Dashboard Filter Actions**, **Highlight Actions**, **Viz in Tooltips**, and **URL Web Actions** on the Netflix dashboard, so that offline stakeholders can explore the data dynamically without needing Tableau Server or authoring tools.

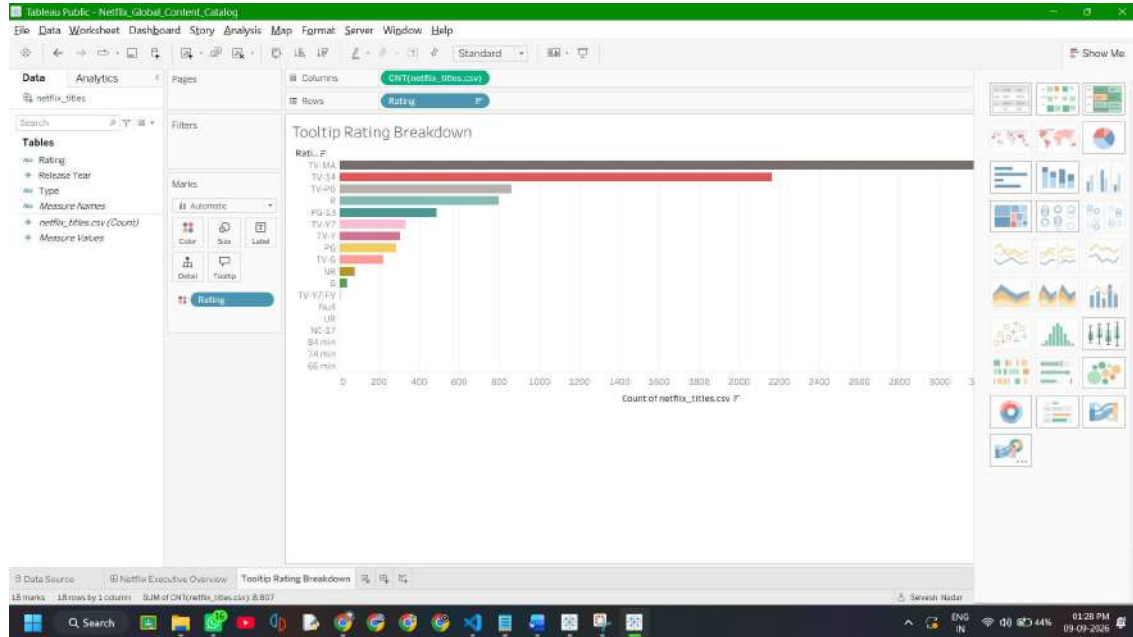
An executive dashboard should not be a static grid of charts. Adding interactive actions allows users to click a category to filter time-series trends across other charts, hover over a summary bar to see a detailed micro-chart embedded in the tooltip, and launch contextual external searches directly from data marks.

Step 1: Create the Detail Sheet for Viz in Tooltip

A Viz in Tooltip dynamically renders an embedded secondary chart inside the hover tooltip of a primary chart.

1. Create a new worksheet named: **Tooltip Rating Breakdown**.
2. Drag **Rating** onto the **Rows** shelf.
3. Drag **netflix_titles.csv (Count)** onto the **Columns** shelf.
4. Drag **Rating** onto the **Color** tile on the Marks card.
5. Sort the bars in descending order.

Data Visualization with Power BI and Tableau

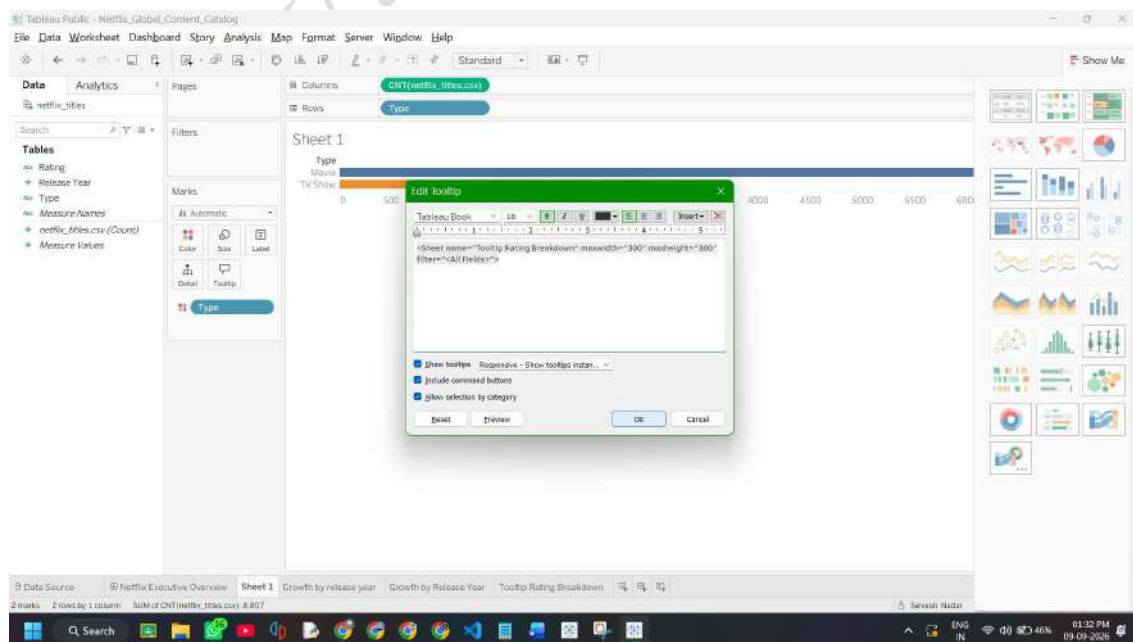


Step 2: Embed the Viz into the Main Tooltip

1. Return to your **Content Type Split** worksheet.
2. On the **Marks** card, click the **Tooltip** tile.
3. In the Edit Tooltip dialog:
 - o Delete default text if desired, then click **Insert** in the top-right corner of the editor.
 - o Select **Sheets --> Tooltip Rating Breakdown**.
 - o Tableau inserts the code tag:

```
<Sheet name="Tooltip Rating Breakdown" maxwidth="300"
maxheight="300" filter="All Fields">
```

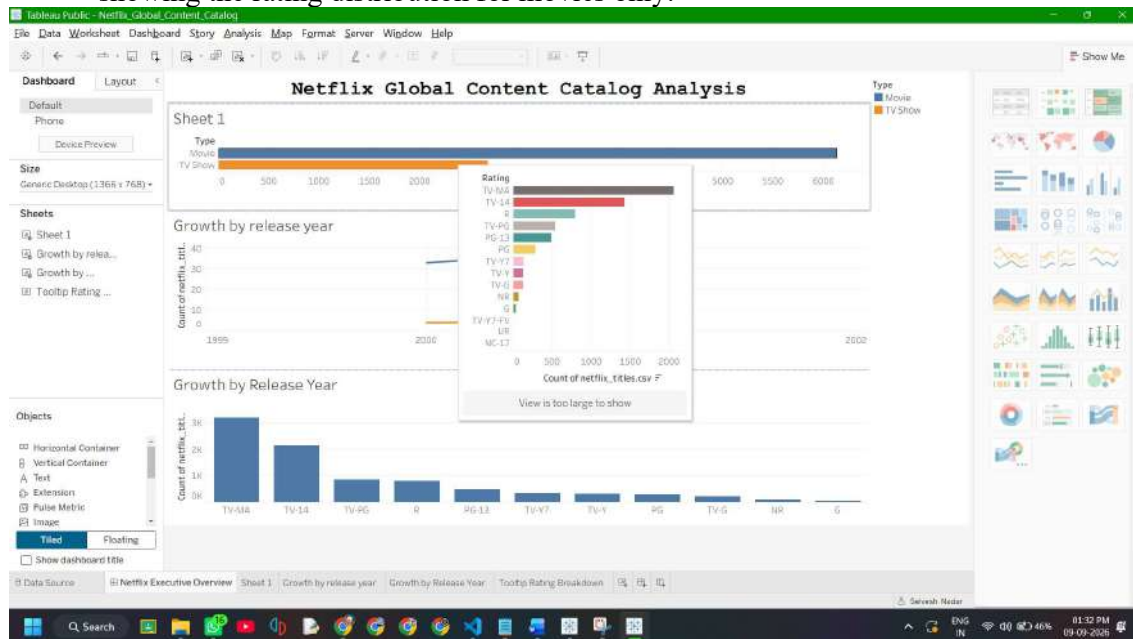
4. Click **OK**.



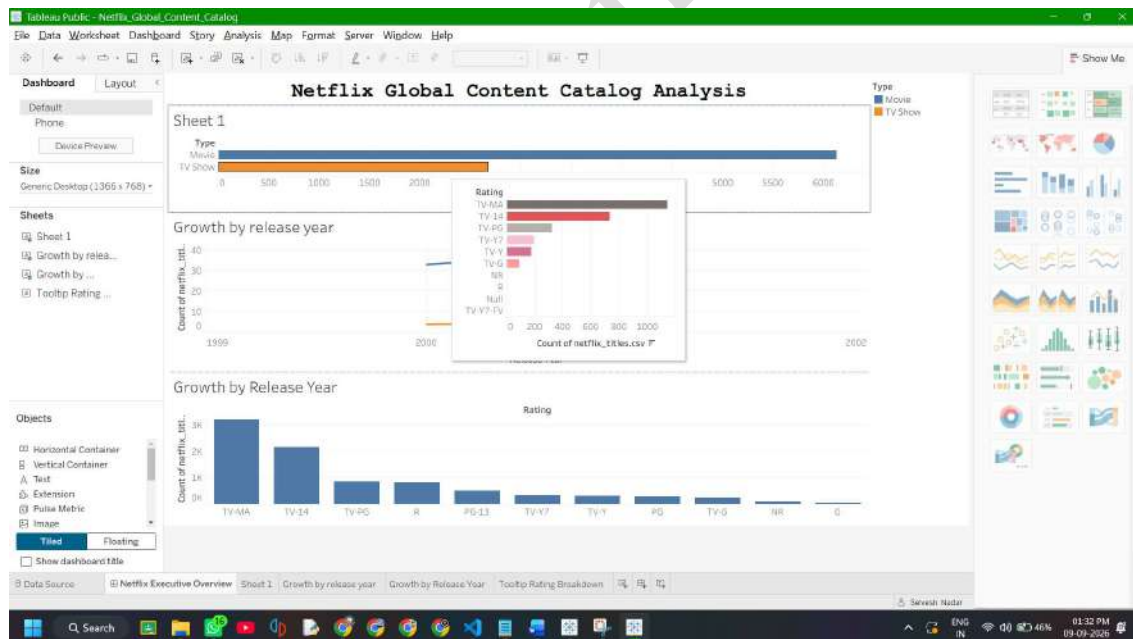
Data Visualization with Power BI and Tableau

5. Switch back to your **Netflix Executive Overview** dashboard and test it:

- Hover your mouse over the **Movie** bar: you will see a mini-bar chart inside the tooltip showing the rating distribution for movies only.



- Hover over the **TV Show** bar: the mini-bar chart instantly changes to show TV show ratings (TV-MA, TV-14, etc.).

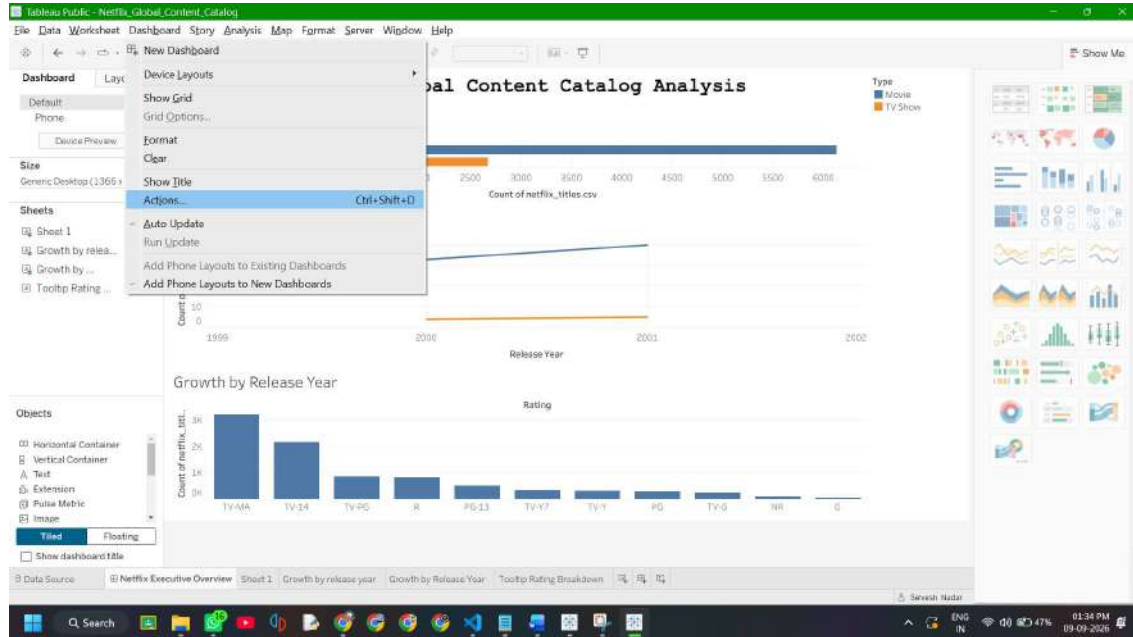


Step 3: Configure Dashboard Filter Actions (Cross-Filtering)

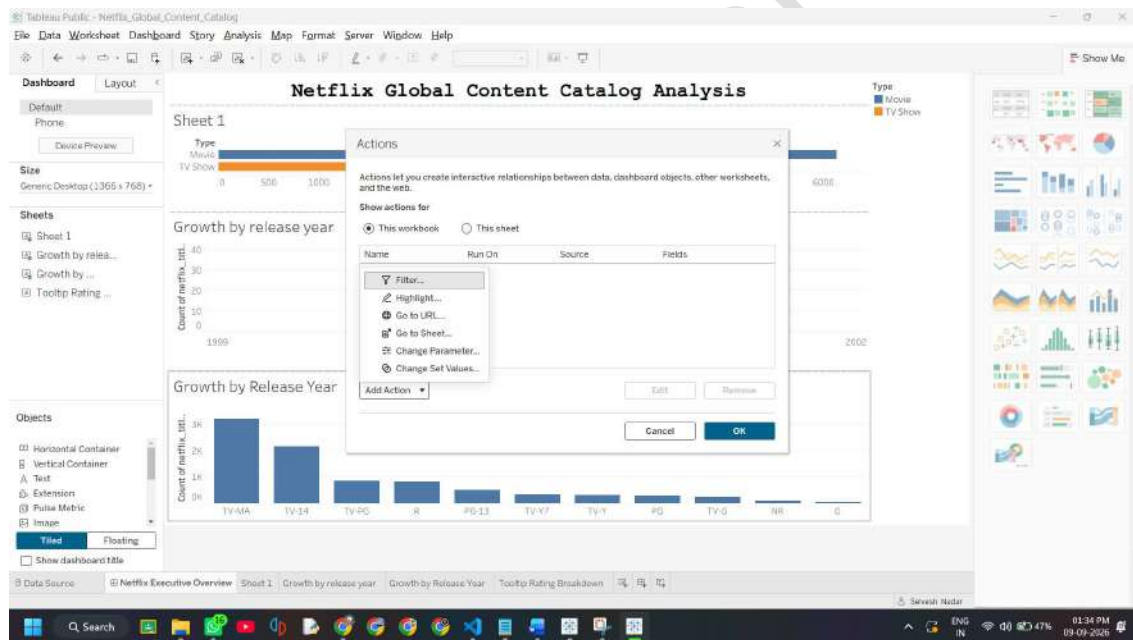
Clicking a mark in one chart will filter the other charts on the dashboard.

1. Open your **Netflix Executive Overview** dashboard.
2. From the top menu bar, select **Dashboard --> Actions...** (or press Ctrl + Shift + A).

Data Visualization with Power BI and Tableau



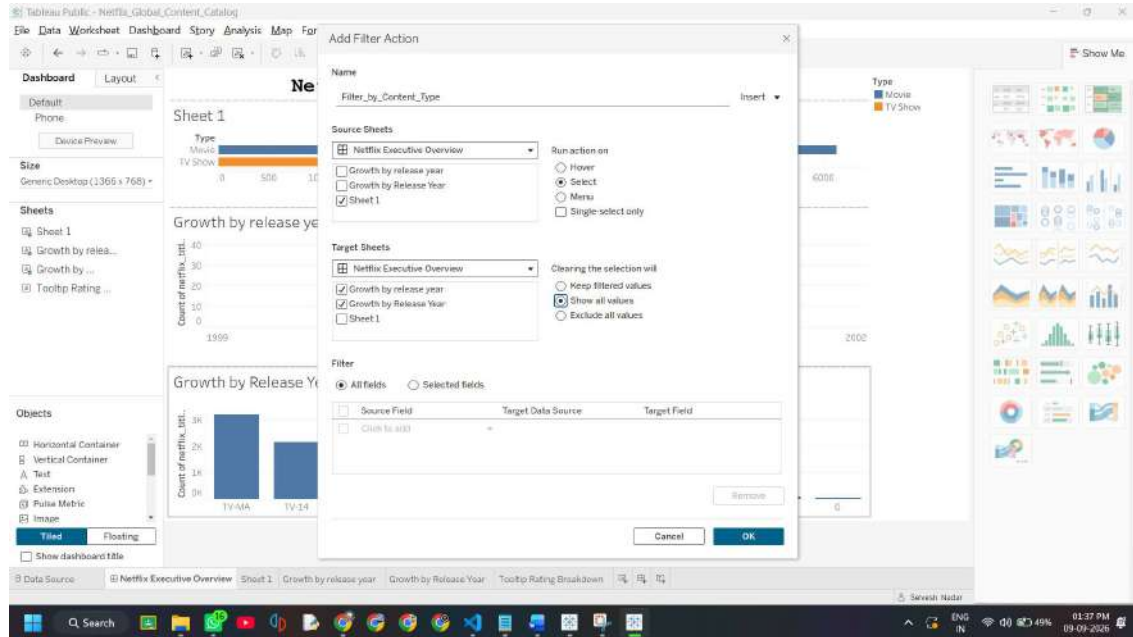
3. In the Actions dialog, click **Add Action** --> select **Filter...**



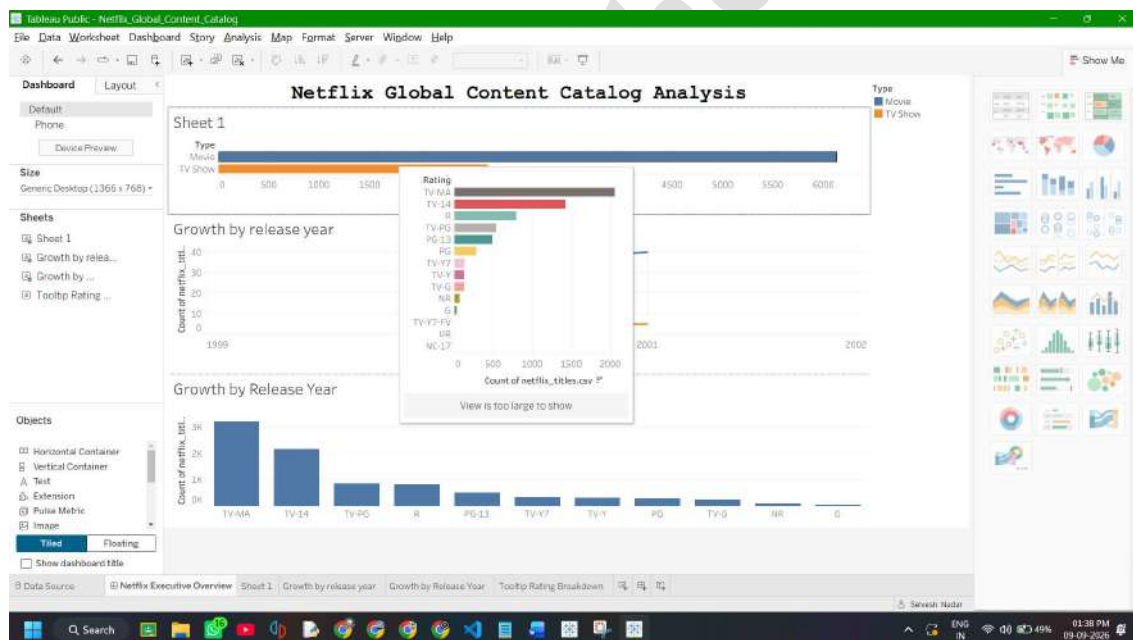
4. Configure the Filter Action:

- **Name:** Filter_by_Content_Type
- **Source Sheets:** Check **Content Type Split.** (Sheet 1)
- **Run action on:** Select **Select** (triggers on click).
- **Target Sheets:** Check **Growth by Release Year** and **Top 10 Ratings**.
- **Clearing the selection will:** Select **Show all values**.
- **Target Filters:** Leave as **All Fields**.
- **NOTE:** Name of Sheets can be different.

Data Visualization with Power BI and Tableau



5. Click **OK**, then click **OK** again.
6. **Test the Filter:** Click the **Movie** bar; the line chart instantly updates to show release year trends exclusively for movies. Click it again to clear the filter.

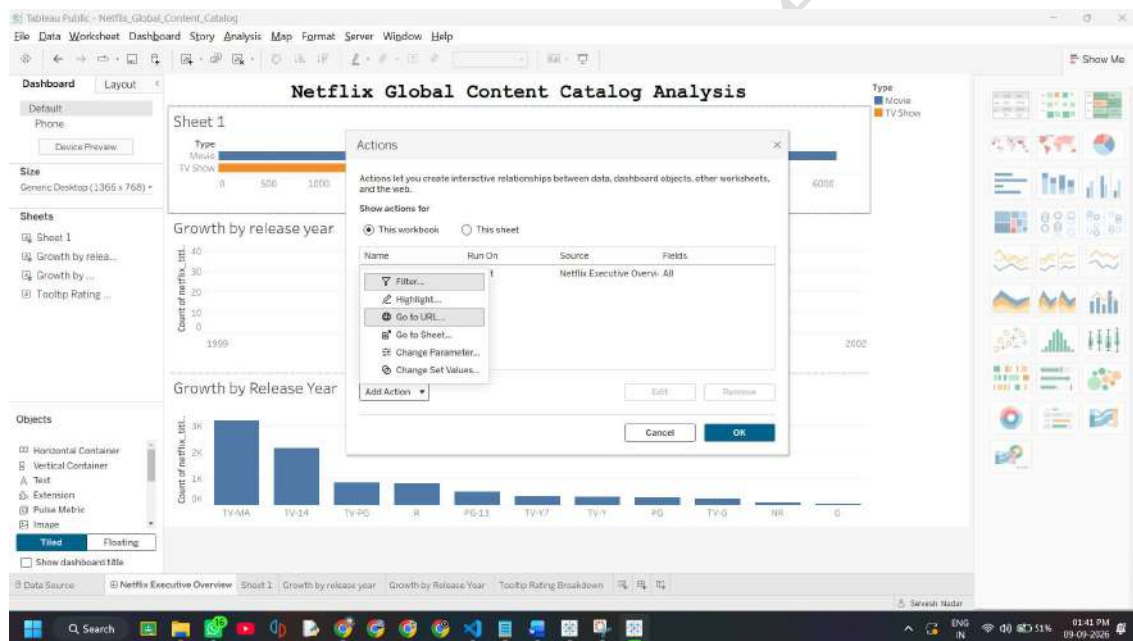
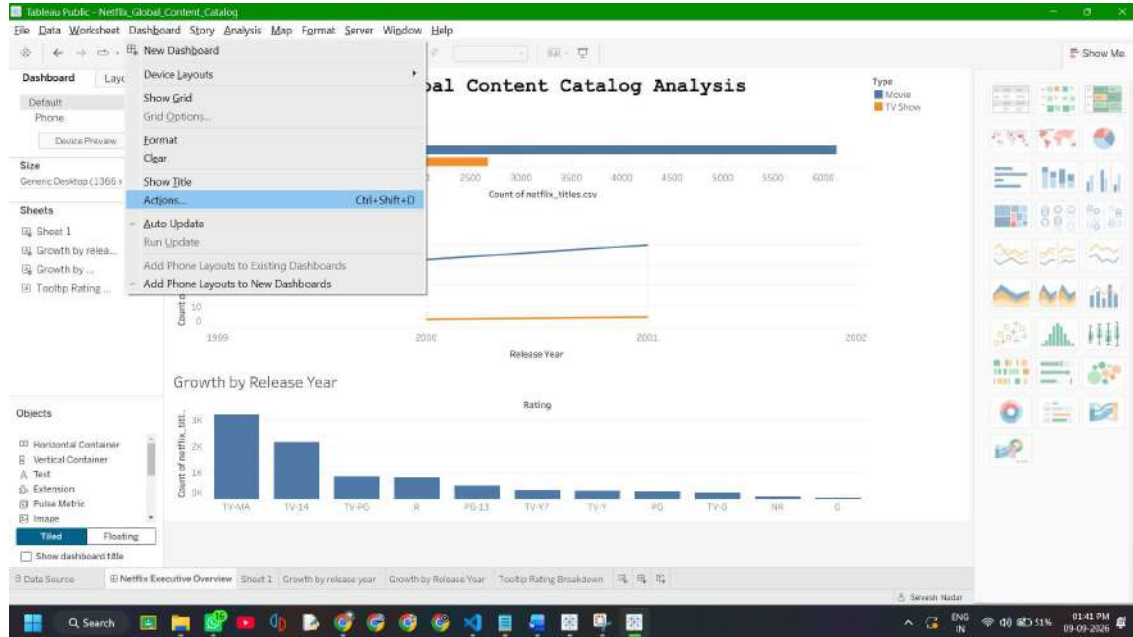


Step 4: Add a Contextual Web URL Action

URL Actions allow users to click a mark and open an external web page (e.g., searching Google or IMDb for a selected title or genre).

1. Go to **Dashboard --> Actions...** --> click **Add Action** --> select **Go to URL....**

Data Visualization with Power BI and Tableau



2. Configure the URL Action:

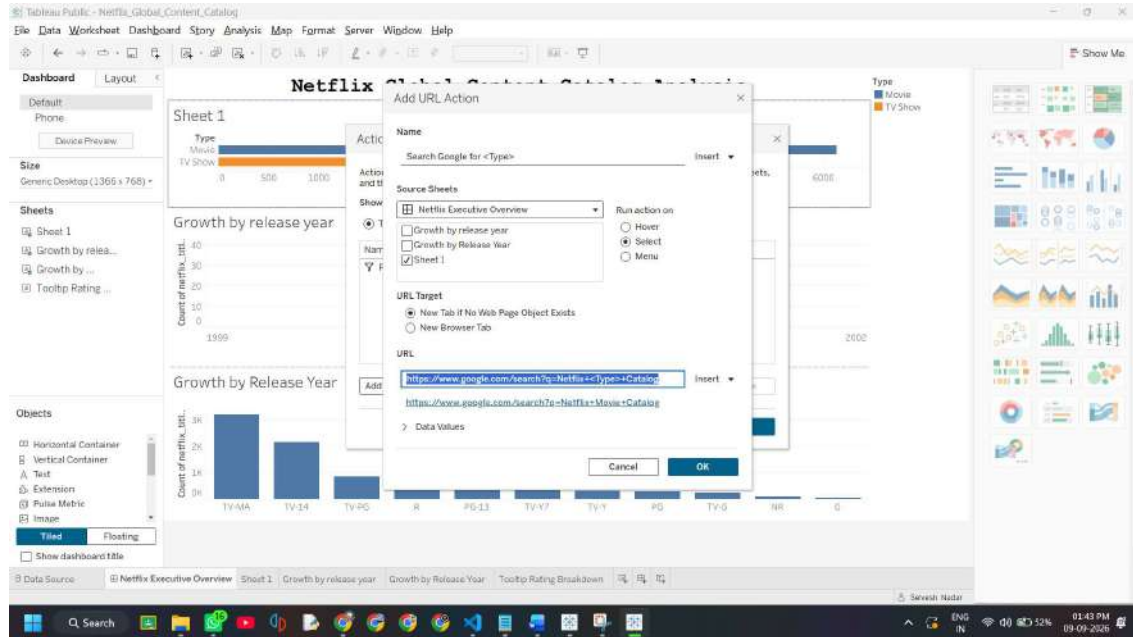
- **Name:** Search Google for <Type>
- **Source Sheets:** Check **Content Type Split**.
- **Run action on:** Select **Menu** (adds a clickable link inside the tooltip).
- **URL:** Enter:

<https://www.google.com/search?q=Netflix+<Type>+Catalog>

- (Click the small arrow next to URL to insert the <Type> dimension dynamically).

3. Click OK, then click OK to close.

Data Visualization with Power BI and Tableau



4. Hover over any bar and click the hyperlink menu item to verify that the browser opens the dynamic search query.

F: Manage Permissions and Access Control

Version & Environment Note: In standalone, unactivated trial, or local Tableau Desktop environments without an active connection to Tableau Server/Cloud, built-in user directories and the bottom-right "*Filter as User*" menu are disabled. To fulfill the core objective of **Row-Level Security (RLS)** and **Role-Based Access Control (RBAC)** locally, we simulate user personas and security clearance using dynamic parameters and conditional logic.

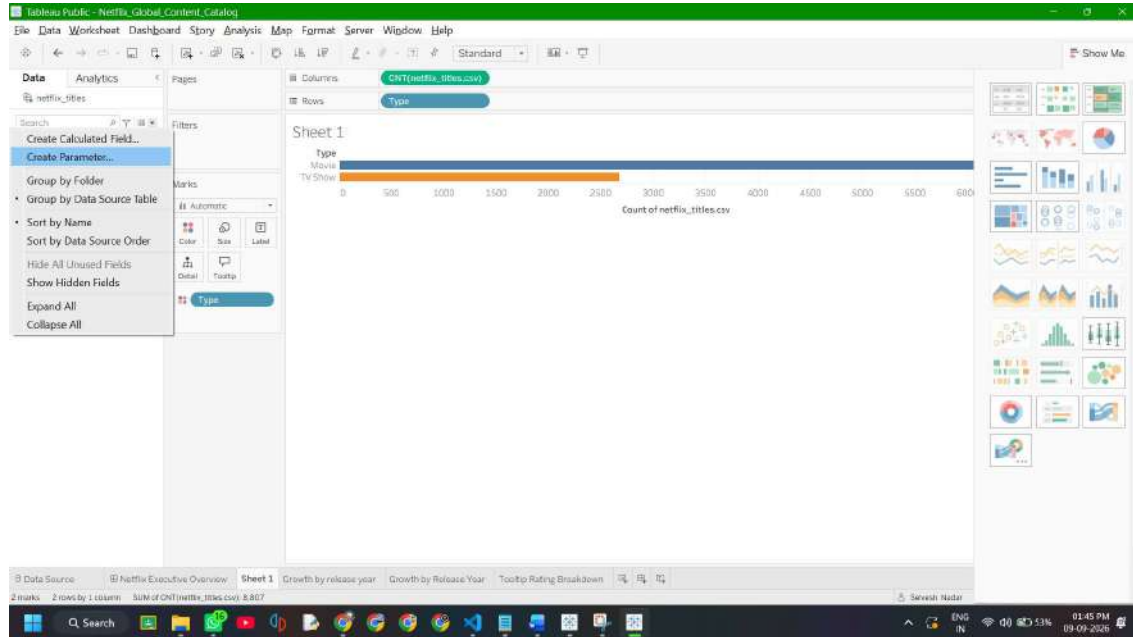
Objective: Build and test a local **Row-Level Security (RLS)** filter that restricts dataset visibility based on user roles and regional clearance.

In enterprise reporting, sensitive data must be filtered based on who is viewing the report (e.g., a regional analyst should only see local market data, while executive leadership sees the complete global catalog). Implementing security logic at the calculation and data source layer ensures single-workbook scalability without unauthorized data exposure.

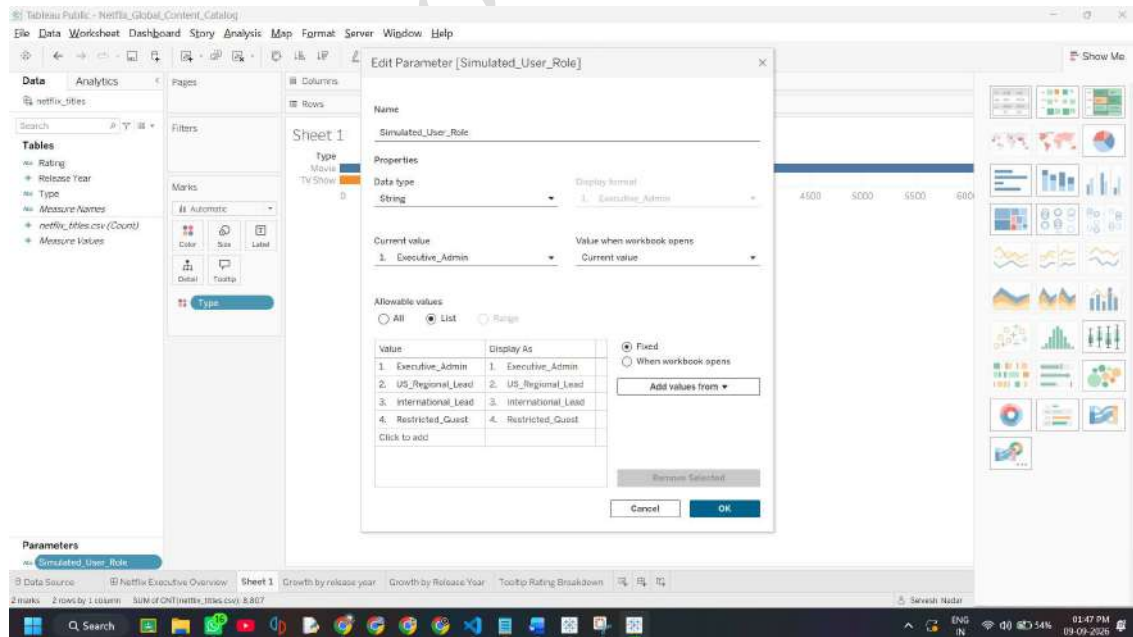
Step 1: Create the User Persona Simulation Parameter

1. In the **Data Pane** on the left, click the small dropdown arrow next to the search bar (or right-click an empty space) --> select **Create Parameter**....

Data Visualization with Power BI and Tableau

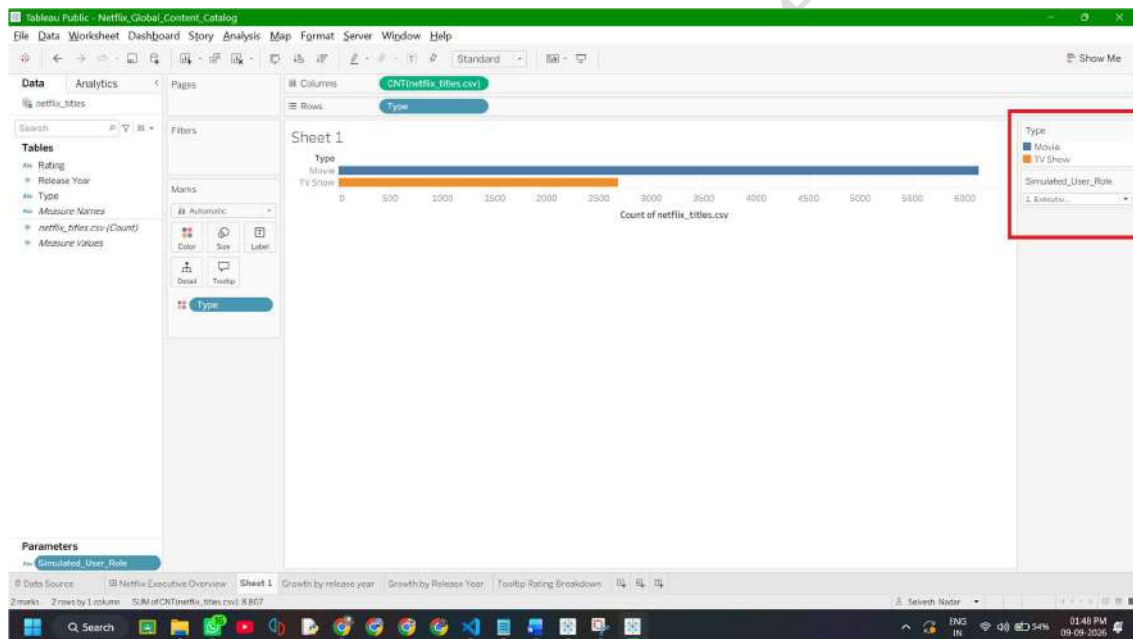
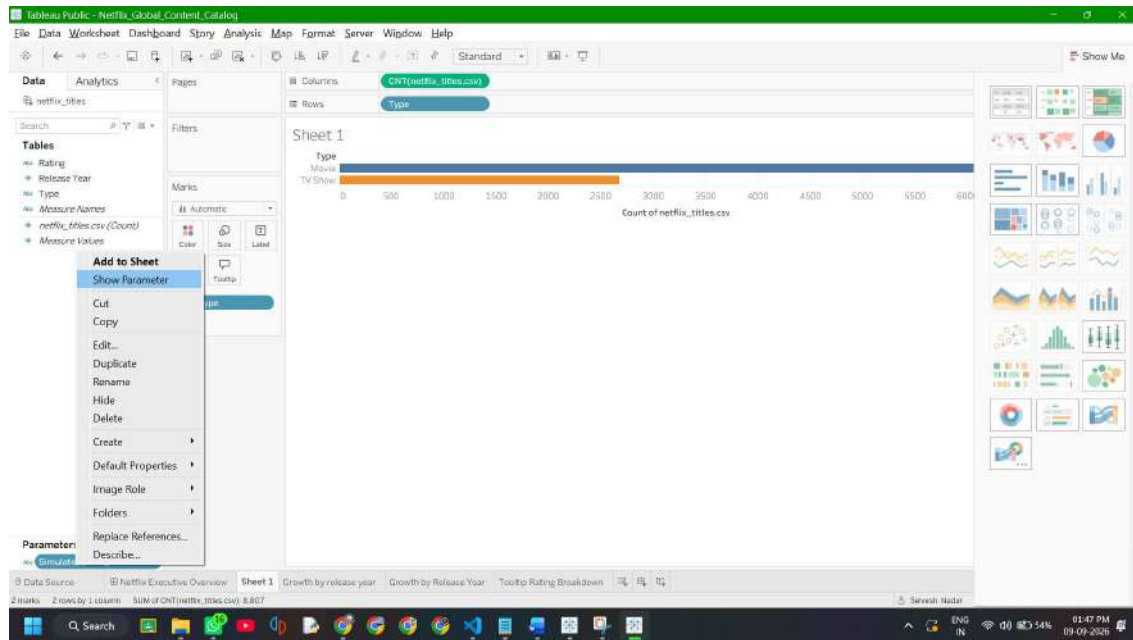


2. Configure the parameter properties:
 - **Name:** Simulated_User_Role
 - **Data type:** String
 - **Allowable values:** Choose List
 - Add four simulated corporate identities:
 1. Executive_Admin
 2. US_Regional_Lead
 3. International_Lead
 4. Restricted_Guest
 - **Current value:** Set to Executive_Admin



3. Click OK.
4. In the Data Pane under **Parameters**, right-click **Simulated_User_Role** --> select **Show Parameter**.

Data Visualization with Power BI and Tableau



Step 2: Write the Access Control Calculated Field

1. Right-click an empty space in the Data Pane --> select **Create Calculated Field...**
2. Configure the field:

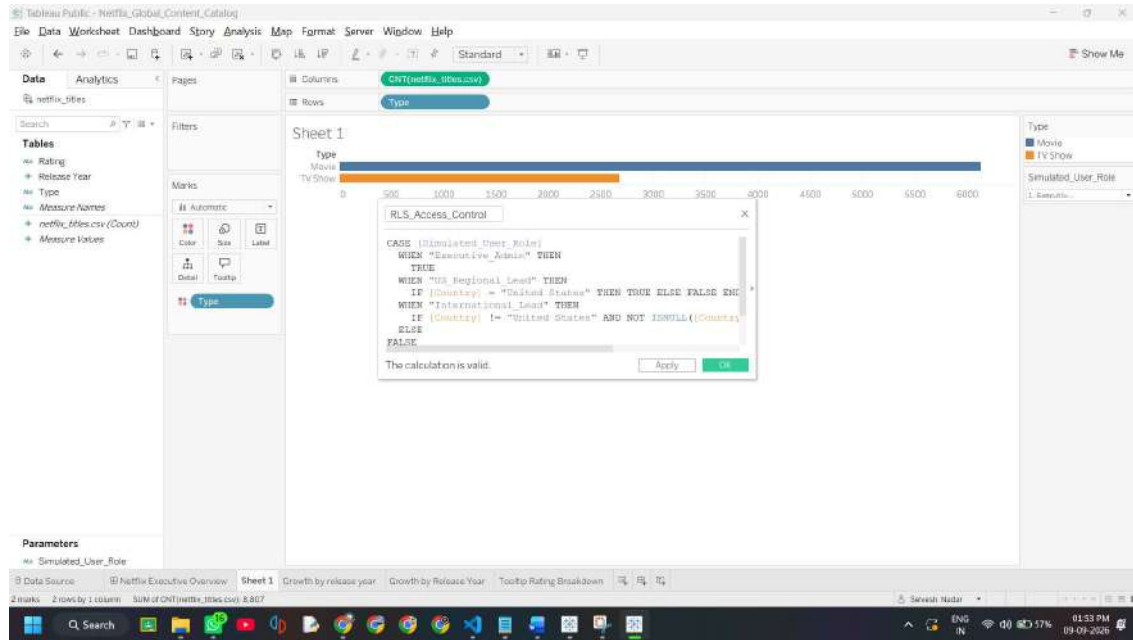
- **Field Name:** RLS_Access_Control
- **Tableau Formula:**

```
CASE [Simulated_User_Role]
  WHEN "Executive_Admin" THEN
    TRUE
  WHEN "US_Regional_Lead" THEN
    IF [Country] = "United States" THEN TRUE ELSE FALSE END
  WHEN "International_Lead" THEN
    IF [Country] != "United States" AND NOT ISNULL([Country])
    THEN TRUE ELSE FALSE END
  ELSE
```


Data Visualization with Power BI and Tableau

FALSE

END

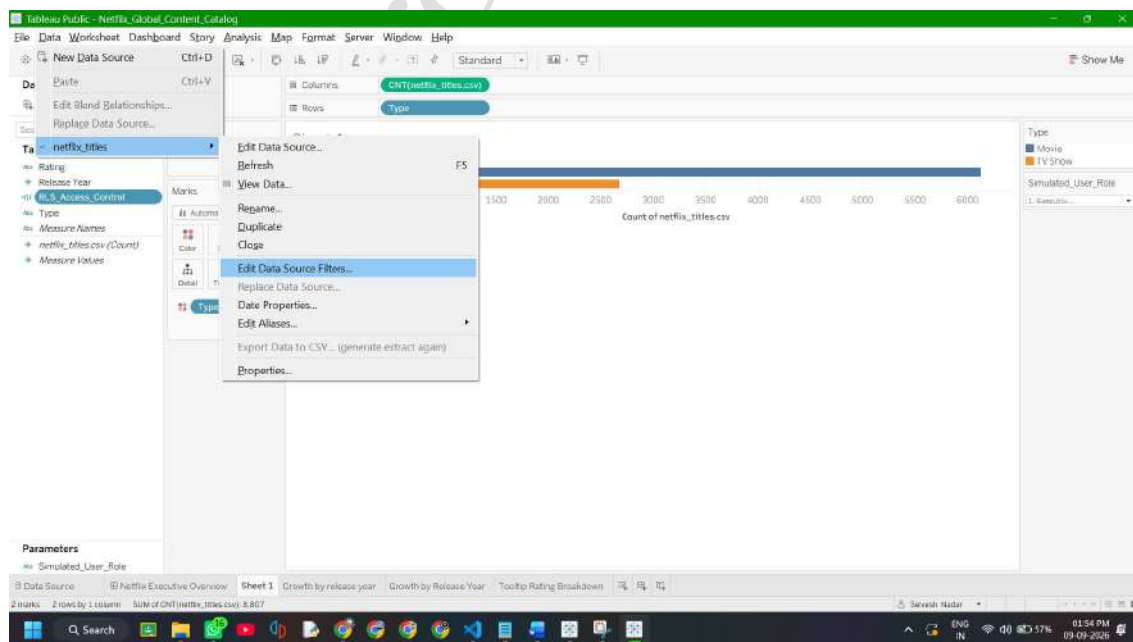


3. Click OK.

Step 3: Enforce Access Control at the Data Ingestion Layer

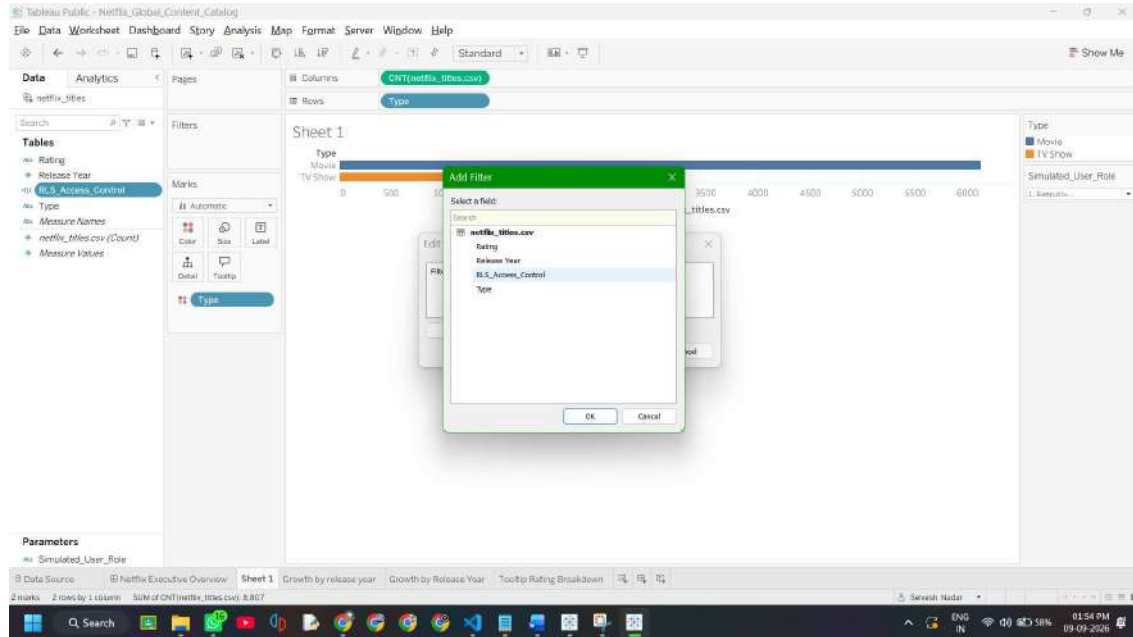
To make this rule tamper-proof across all sheets in the workbook, apply it as a global Data Source Filter.

1. Go to the top application menu bar --> select **Data** --> hover over **netflix_titles** --> select **Edit Data Source Filters**....

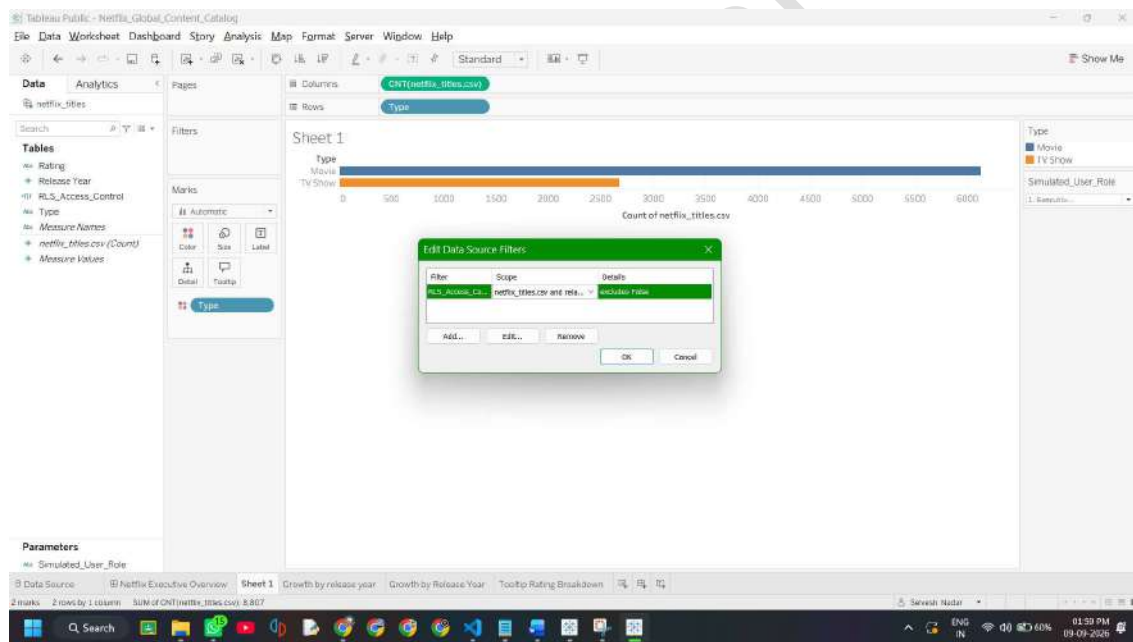


2. In the Data Source Filters dialog, click **Add**....
3. Select **RLS Access Control** from the field list --> click **OK**.

Data Visualization with Power BI and Tableau



4. In the filter dialog, check the box for **True** --> click **OK**, then click **OK** again.



Step 4: Test and Verify Role-Based Permissions

Switch the **simulated_user_role** parameter dropdown on your dashboard to observe real-time access enforcement:

- **Select Executive_Admin:** The entire global catalog of 8,807 titles across all countries is accessible.
- **Select US_Regional_Lead:** The dashboard automatically filters to show only the 2,818 United States titles.
- **Select International_Lead:** All US titles are masked, leaving only non-US international content.

Data Visualization with Power BI and Tableau

- **Select Restricted_Guest:** The access filter returns `FALSE` for all rows, blanking out the view entirely (zero data exposure).

G: Evaluate and Optimize Sharing Strategies

Objective: Evaluate the four primary offline/local distribution methods for Tableau assets—**Packaged Workbooks** (.twbx), **Standard Workbooks** (.twb), **Vector PDFs**, and **Static Images** (.png/.emf) and apply optimization techniques to minimize file size, remove redundant data, and ensure peak rendering performance.

In environments without an active Tableau Server or cloud repository, sharing data products requires sending standalone files via email, shared network drives, or embedding them into executive presentations. Choosing the wrong local format or failing to optimize data extracts leads to bloated file attachments, broken data connection paths, and sluggish user interactions.

Step 1: Comparative Evaluation of Local Distribution Strategies

Distribution Method	Output File Format	Recipient Requirements	Interactivity Level	Best Use Case	Key Vulnerability / Trade-off
Packaged Workbook	.twbx	Tableau Desktop or free Tableau Reader	High (Full cross-filtering, tooltips, parameters)	Sending interactive dashboards to colleagues for standalone exploration	Embeds the entire dataset inside the file; security risk if underlying data is sensitive
Standard Workbook	.twb	Tableau Desktop + Access to original data path	High (Full authoring and filtering)	Developer version control and local script repositories	Contains only XML metadata and visualization code; will show broken links if the CSV moves
Vector Document	.pdf	Any PDF reader (Adobe, Web Browser)	None (Static printout)	Formal board packs, audited regulatory submissions, executive briefings	Cannot drill down or interact; multi-page scaling must be managed manually

Data Visualization with Power BI and Tableau

Distribution Method	Output File Format	Recipient Requirements	Interactivity Level	Best Use Case	Key Vulnerability / Trade-off
High-Res Graphic	.png / .emf	Slide software (PowerPoint, Keynote)	None (Static snapshot)	Slide decks, email summaries, internal wiki articles	Fixed resolution; tooltip details and hidden marks are lost

Step 2: Practical Optimization Checklist for Local Dashboards

To ensure local workbooks open fast and remain under email file size limits (e.g., $< 25 \text{ MB}$), apply these three optimizations directly in Tableau Desktop:

1. Hide All Unused Fields (Data Extract Pruning)

By default, Tableau extracts every column from the source file. Removing unreferenced columns significantly shrinks the .hyper and .twbx footprint.

1. Go to the **Data Pane** on the left.
2. Click the small drop-down arrow next to the search bar --> select **Hide All Unused Fields**.
3. Right-click your data source (netflix_titles) --> select **Extract Data...** --> click **Extract** to regenerate a trimmed extract.
4. We have already done this step earlier.

2. Eliminate Calculation Overheads

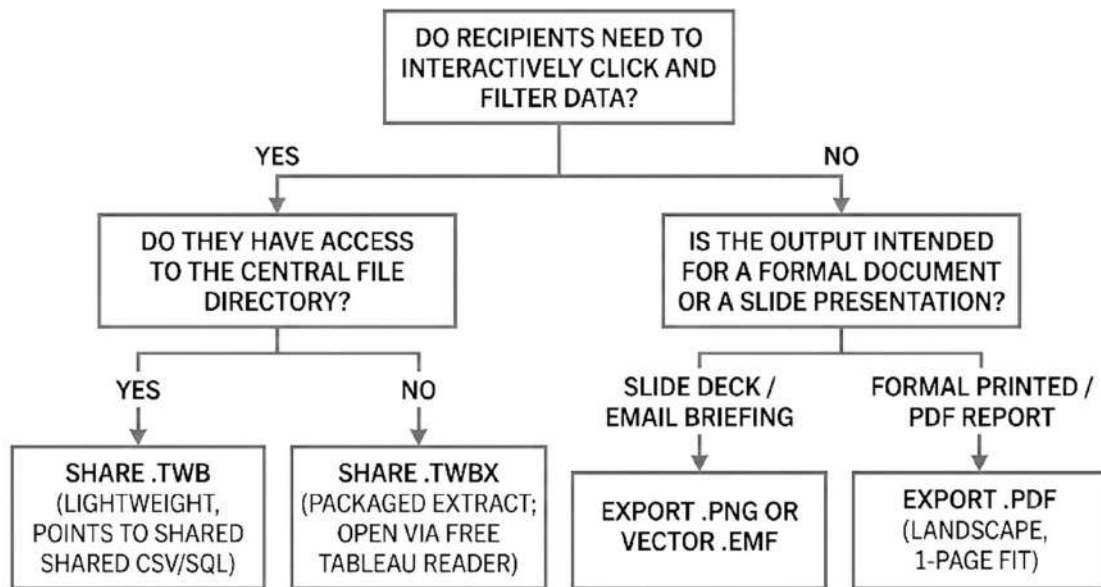
- Convert row-level string manipulations into boolean or integer flags where possible.
- Avoid nested IF-ELSE calculations on high-cardinality text fields inside the view; use native **Grouping** or **Sets** instead.

3. Standardize Layout Sizing for Presentations

- For files exported to slide decks or opened on varying monitors, set the dashboard size to **Fixed Size** (e.g., 1366×768 or 1920×1080) rather than *Automatic*.
- *Why*: Fixed sizing forces Tableau to cache the render pass once, preventing re-rendering calculations every time a recipient resizes their window.

Step 3: Local Deployment Decision Workflow

Data Visualization with Power BI and Tableau



~~~~~ End of lesson 12 ~~~~~